

**HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY AND
EDUCATION
FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING
DEPARTMENT OF COMPUTER AND COMMUNICATION
ENGINEERING**



FINAL COURSE PROJECT

**SMART WATER TANK: LEVEL AND
FLOW MONITORING WITH
AUTOMATIC PUMP CONTROL**

MAJOR IN COMPUTER ENGINEERING

Students: Trịnh Trần Quốc Bảo
Student ID: 24119016
Đặng Cao Dương
Student ID: 24119024
Huỳnh Lê Thành Nhân
Student ID: 24119068
Nguyễn Đoàn Hoàng Ngân
Student ID: 24119064
Tô Ngọc Thảo Uyên
Student ID: 24119098

Supervisor: Dr. HUYNH THE THIEN

HO CHI MINH CITY, 10/2026

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Throughout this final project for the course IoT Foundations and Applications, our group received careful guidance and valuable technical feedback from Dr. Huynh The Thien. His lectures on layered architecture, application layer messaging and time series data infrastructure provided the theoretical basis for most of the design decisions in the system we built.

We also thank the Department of Computer and Communication Engineering for granting us laboratory access, which allowed us to complete the hardware prototype and carry out the measurements reported here.

We would like to express our sincere and deepest gratitude.

Project group
(Signature and full name)

DECLARATION OF AUTHORSHIP

We declare that the entire content presented in this report is the result of our own research and implementation. The firmware, the backend service and the monitoring interface were written by us. The experimental data were collected directly on the physical prototype that we assembled. Every item taken from the course lectures, technical standards and scientific literature is cited at the point of use and listed in the references.

Project group
(Signature and full name)

ABSTRACT

This project builds a closed loop smart water tank for Project 08: it measures water level and flow, drives a pump automatically, accumulates consumed volume, detects and classifies faults, and can be supervised from anywhere on the Internet. The system spans the layered model of the Internet of Things: an ESP32 with an ultrasonic level sensor, two turbine flow sensors, a Hall current sensor and two float switches; MQTT over Wi-Fi to a Mosquitto broker; a FastAPI service with a SQLite time series store; and a web dashboard published through an HTTPS tunnel, where anyone may watch and only a password authenticated administrator may control.

The central design decision is to place the entire control law and every safety rule inside the edge microcontroller. Level regulation therefore keeps working while the network is down, and a ring buffer replays telemetry when it returns. The controller is a seven state machine with a 30 to 70 percent hysteresis band and minimum on and off times; eleven fault rules, a single guard function through which every pump start must pass, and independent overflow layers ensure that manual commands cannot bypass safety. Level readings pass a robust filter chain (lower quartile of a fifteen ping window, interquartile spread gate, physical rate checks and an alpha-beta filter), backed by a tank model that bridges intervals without a trusted reading, and a pump health classifier separates hydraulic from electrical pump faults using current and level evidence. A rolling baseline over 48 half hour slots detects abnormal consumption on the backend.

On the bench, every automatic fill of the control session stopped between 70.1 and 70.8 percent with at most 1.85 percent overshoot, the pump started 12.9 times per hour with no relay chatter, no false alarm was raised in 84 minutes, and pumped volume was estimated with a mean error of -5.2 percent against a geometric reference. Calibrated against a ruler and the pumped volume, the level reads within 0.68 centimetres root mean square outside one ultrasonic blind zone. On the final firmware the tank kept filling through a two minute network outage and stopped itself at 70.7 percent, 4 of 129 messages were lost, and commands were confirmed with a median round trip of 29 milliseconds. The report states openly where the prototype falls short: the flow sensors are fitted and the inlet sensor is calibrated, but control and accounting still use flow derived from the level; the no current fault was not injected physically; and the MQTT path inside the laboratory is not encrypted.

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List of Abbreviations

The following abbreviations are used throughout this report.

Abbreviation	Definition
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
QoS	Quality of Service
LWT	Last Will and Testament
ACL	Access Control List
TLS	Transport Layer Security
REST	Representational State Transfer
API	Application Programming Interface
JSON	JavaScript Object Notation
IQR	Interquartile Range
mDNS	Multicast Domain Name System
RTT	Round Trip Time
HTTPS	Hypertext Transfer Protocol Secure
TSDB	Time Series Database
ADC	Analog to Digital Converter
SAR	Successive Approximation Register
PWM	Pulse Width Modulation
GPIO	General Purpose Input Output
MCU	Microcontroller Unit
FSM	Finite State Machine
ISR	Interrupt Service Routine
NVS	Non-Volatile Storage
EMF	Electromotive Force
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
EWMA	Exponentially Weighted Moving Average
CUSUM	Cumulative Sum

Chapter 1

INTRODUCTION AND BACKGROUND

1.1 Motivation and problem statement

Household water management in Vietnam still relies largely on a mechanical float placed inside the storage tank. The device is cheap and durable, but it solves exactly one problem: cutting the pump when the tank is full. The owner has no idea how much water was consumed, cannot notice a pipe leak until the bill rises unexpectedly, and has no way of knowing that the pump is running dry until it has already failed.

The problem carries every characteristic of a cyber physical system, meaning a system in which computation, communication and a physical process are tied together through feedback loops [1]. A physical quantity is measured, digitised, transported to a data infrastructure, turned into a decision, and finally acted back upon the same physical process through an actuator.

From an engineering point of view the tank is a nonlinear plant carrying a hard safety constraint. Discharge flow depends on the square root of the water column height, so the plant response changes with the current level. At the same time there is one constraint that may never be violated under any circumstance, including a manual command issued by an operator: the tank must not overflow. Having to guarantee control quality and an absolute safety constraint at the same time is what lifts this project above a simple exercise in wiring a sensor to a display.

The system must therefore address four groups of problems: measuring level and flow with an error budget established experimentally rather than copied from a datasheet; holding the level inside a desired band without letting the pump chatter around the switching threshold; detecting abnormal hydraulic operating modes; and providing remote supervision through a time series store and a real time interface. Overarching all of them is a single requirement that shapes the whole architecture: essential level control must keep working when the network link is lost.

1.2 Scope and working method

The system is built at laboratory prototype scale with one sensing node, a tank with a 1.24 litre working volume and a low voltage direct current pump. The entire electrical side operates at safe extra low voltage. Distributed multi node deployment, machine learning based consumption forecasting and integration with a commercial cloud platform are outside the scope of this work.

The group organised the work along the V model introduced in the project management part of the course [1], in which every decomposition step on the descending branch has a matching verification step on the ascending branch: requirement analysis pairs with full system acceptance, architectural design pairs with integration testing, protocol specification pairs with protocol level testing, and circuit design pairs with unit testing of each individual sensor.

One methodological decision deserves emphasis. The group specified the complete MQTT topic tree, the message payloads, the set of API endpoints and the database schema before writing a single line of code. That specification acted as an internal contract, which allowed the firmware branch, the backend branch and the interface branch to advance in parallel instead of waiting on one another. While components had not yet arrived, a simulator publishing messages that conform to the agreed specification let the other two branches be developed and tested in full.

1.3 Requirements of the assignment

The assignment, Project 08, asks for a smart water tank that monitors consumption and controls its pump automatically. Table 1.3.1 restates each requirement in the words of the brief and points to where the report answers it. The third column is the group's own assessment of how completely the requirement is done, because a requirement that is only partly met is better stated than hidden.

Table 1.3.1: Requirements of Project 08 and where they are addressed.

Requirement	Where	Status
ESP32 class controller with tank level and flow sensors; pressure and pump current optional	Sections 2.3, 3.2, 5.2	Partly validated: both fitted and read, inlet calibrated; control and volume use level derived flow
Low voltage pump through a relay or MOSFET driver with inductive protection	Section 2.4	Done
Prevent overflow and behave safely on failed or implausible sensors	Sections 3.4, 4.7	Done, seven independent layers
Instantaneous flow and accumulated volume	Sections 3.2, 5.2	Done, from level; mean error -5.2%
Publish telemetry, actuator state, faults and availability over MQTT	Sections 2.5 to 2.8	Done
Time series storage, REST API, dashboard with level, flow, daily volume, pump state, alarms and freshness	Sections 4.1 to 4.4	Done
Automatic control with hysteresis or a state machine and at least one fault rule, such as pump on with near zero flow	Sections 3.3, 3.4	Done, eleven rules
Manual mode cannot bypass safety; commands acknowledged with confirmed state	Sections 3.5, 2.7	Done
Local level control keeps working during a network outage	Sections 2.2, 4.7, 5.5	Done
Advanced: leak detection from a rolling baseline, or pump current and flow consistency to classify pump faults	Sections 3.6, 3.7	Both built; baseline still converging
Calibrate level and flow against a physical reference	Section 5.2	Done: ruler and pumped volume, RMSE 0.68 cm; one blind zone
Control response, switching frequency, fault detection delay, volume estimation error	Sections 5.2 to 5.4	Done
Sensor or no flow fault test, network loss and recovery, telemetry and control latency	Sections 5.4, 5.5	Done; outage rerun on final firmware; NO_CURRENT not injected

1.4 Layered architecture and computation placement

An Internet of Things deployment is commonly described by a five layer model made of perception, transport, processing, application and business layers, allowing each stage of signal conversion, transport and storage to be optimised independently [1]. Recommendation ITU-T Y.2060 adds two capability groups that cut across the stack, management and security [2]. Treating security as cross cutting rather than as a separate layer has a direct consequence for our design: it cannot be handled at a single point but must appear in broker authentication, per topic access control, transport encryption and API authorisation. Data flow is likewise split into a southbound path carrying commands down to the hardware and a northbound path carrying telemetry and alerts up to the server [1].

Deciding where a computation belongs is driven by quantitative constraints rather than taste. Edge computation offers deterministic latency and very low bandwidth cost but only short term storage, while cloud computation offers durable storage and analytic capacity at the price of latency from hundreds of milliseconds to several seconds [1]. The resulting rule is that any operation requiring an immediate safe stop must be computed at the edge, whereas trend analysis is routed upward. Cutting the pump on a dry run condition or when the level passes the safety threshold clearly belongs to the first category, which is the direct basis for the decision in Section 2.2.

1.5 Sensing, digitisation and actuation

Faithful reconstruction of a continuous analog signal requires a sampling rate greater than twice its highest frequency component [3]. With the pump actually fitted and the tank base measuring 10 by 10 centimetres, the group measured the fill rate directly rather than deriving it from the datasheet. The level rose 1.2 centimetres in 20 seconds, that is 0.06 centimetres per second, equivalent to 0.36 litres per minute. This is nearly five times slower than the 1.67 litres per minute the JT-DC3L datasheet promises at zero head, the difference being delivery head and tubing resistance. A sampling period of 200 milliseconds therefore satisfies the condition by roughly three orders of magnitude. The group records both figures deliberately: an earlier draft quoted 0.6 centimetres per second from an unchecked estimate, the datasheet implies 0.28, and the bench measures 0.06. Only the last of the three was used to size the timing thresholds of Section 3.3, because a threshold derived from a datasheet the hardware does not obey will fire during normal operation. Even the bench figure is not a constant: on 5 October, with a lower source bucket, the same pump delivered 0.24 litres per minute, a fact that matters again in Section 5.2. The ESP32 digitises voltage using a successive approximation register converter [1, 4], and mapping a continuous range onto finite integer steps introduces an unavoidable loss of precision with a quantisation step of

$$\Delta V = \frac{V_{ref}}{2^N} \quad (1.1)$$

and quantisation noise bounded within $\pm \frac{1}{2}\Delta V$. This error is mathematical in nature and can never

be filtered away in software [1]. At 12 bits and 3.3 volts the step is 0.806 millivolts, a figure used in Section 2.3. The lectures also warn that the transfer curve is severely nonlinear near both ends of the range, which is why our dividers place each signal near the middle of the span.

A microcontroller operating at 3.3 volts with roughly 12 milliamperes per pin cannot perform mechanical work directly, so a power stage with galvanic separation has to be inserted. When current through an inductive load is interrupted abruptly, the collapsing field produces a reverse spike following Lenz law, $V = L di/dt$, which can exceed several hundred volts and destroy the switching device; the standard remedy is a flyback diode placed antiparallel across the load [1].

1.6 Application layer messaging

A standard HTTP header easily exceeds 500 to 1000 bytes, so sending a measurement of a few bytes spends most of the energy budget on framing, and every transaction additionally forces the radio through a complete TCP handshake [5]. MQTT addresses this with a binary frame whose fixed header is two bytes, routed through a broker under a publish and subscribe model that decouples communication in space, time and synchronisation [5, 6].

Three MQTT mechanisms are exploited directly by our system. The three quality of service levels define delivery as at most once, at least once and exactly once, and the lectures caution against defaulting every topic to the highest level because a constrained node will exhaust its memory. The last will mechanism lets the broker publish a pre registered payload by itself once a client vanishes without a clean disconnect, and retained messages make the broker push the current value immediately to a newly subscribing client, eliminating the cold start gap [5]. On the security side, MQTTS carries the same binary frames inside a TLS tunnel on port 8883 instead of the unencrypted port 1883, and the lectures stress that a username and password pair alone is insufficient without an access control list mapping each client to the topics it may read and write [5, 7].

1.7 Data infrastructure and the feedback sync principle

Machine telemetry has three structural invariants: it is append only, indexed primarily by time, and its write to read ratio is very high [8]. A relational engine is built for transactional integrity, so every inserted row forces a rebalancing of an in memory B-tree, whereas a time series engine optimises for tail appends and separates indexed tags from unindexed fields. The lectures also warn about index cardinality explosion when a randomly varying value is placed in a tag position [8], a principle our schema in Section 4.2 follows. The REST style demands client server separation and statelessness, favours nouns over verbs, and asks the designer to evaluate safety $f(s) = s$ and idempotency $f(f(s)) = f(s)$ [9, 10]; on the microcontroller side, firmware must forbid dynamic allocation inside the loop and reserve a fixed buffer at startup [10, 11].

The single idea with the largest influence on our interface design is the feedback sync principle derived

from the device shadow model [8]. The anti pattern to avoid is a dashboard toggle that flips to the on state the instant the user presses it, before anything is known about whether the real machine started. The correct architecture splits the path: the user action maps to an outgoing command topic while the indicator stays unchanged until an explicit acknowledgement arrives from the device. This is the basis for the mechanism described in Section 2.7.

1.8 On off control with hysteresis

The material in this section does not appear in the course slides. The group added it from control engineering literature and states the sources here explicitly, so that it is clearly distinguished from the knowledge taken from the lectures.

On off control, also called two position control, is the simplest form of feedback control, in which the actuator takes one of exactly two states [12]. If the two thresholds coincide, even the smallest measurement noise around that point makes the actuator flip repeatedly, causing rapid wear and repeated inrush events. The standard remedy is to separate them into a hysteresis band inside which the state is held unchanged [13]:

$$u(t) = 1 \text{ if } h < h_L; \quad u(t) = 0 \text{ if } h > h_H; \quad u(t) = u(t^-) \text{ if } h_L \leq h \leq h_H \quad (1.2)$$

The width $h_H - h_L$ is a direct trade off parameter: a wider band lowers the switching frequency but widens the level excursion.

For diagnosis, the group applies the principle of checking consistency between two independent signal sources, following established fault diagnosis theory [14]. Slow leaks belong to the class of small persistent shift detection, for which the cumulative sum method of Page is far more sensitive than a fixed threshold because it accumulates small deviations over time [15].

To select control parameters before hardware existed, the group built a dynamic model from the volume balance $dV/dt = Q_{in} - Q_{out} - Q_{leak}$ with $h = V/A$, where discharge follows Torricelli law $Q_{out} = C_d A_o \sqrt{2gh}$ [16]. The square root dependence makes the plant nonlinear, since discharge at a full tank is appreciably faster than near empty. This is why the hysteresis parameters had to be swept in simulation, as described in Section 5.3.

Chapter 2

SYSTEM ARCHITECTURE, HARDWARE AND PROTOCOL

2.1 Overall architecture

The system consists of four blocks in series: an edge node built around an ESP32 with its sensors and one actuator; an MQTT broker acting as a content based message router; a backend service comprising an ingestion process, a database and an API server; and a web dashboard. A fifth, optional element is an outbound tunnel that publishes the dashboard on a public HTTPS address so that the tank can be supervised from anywhere with Internet access, not only from inside the laboratory network.

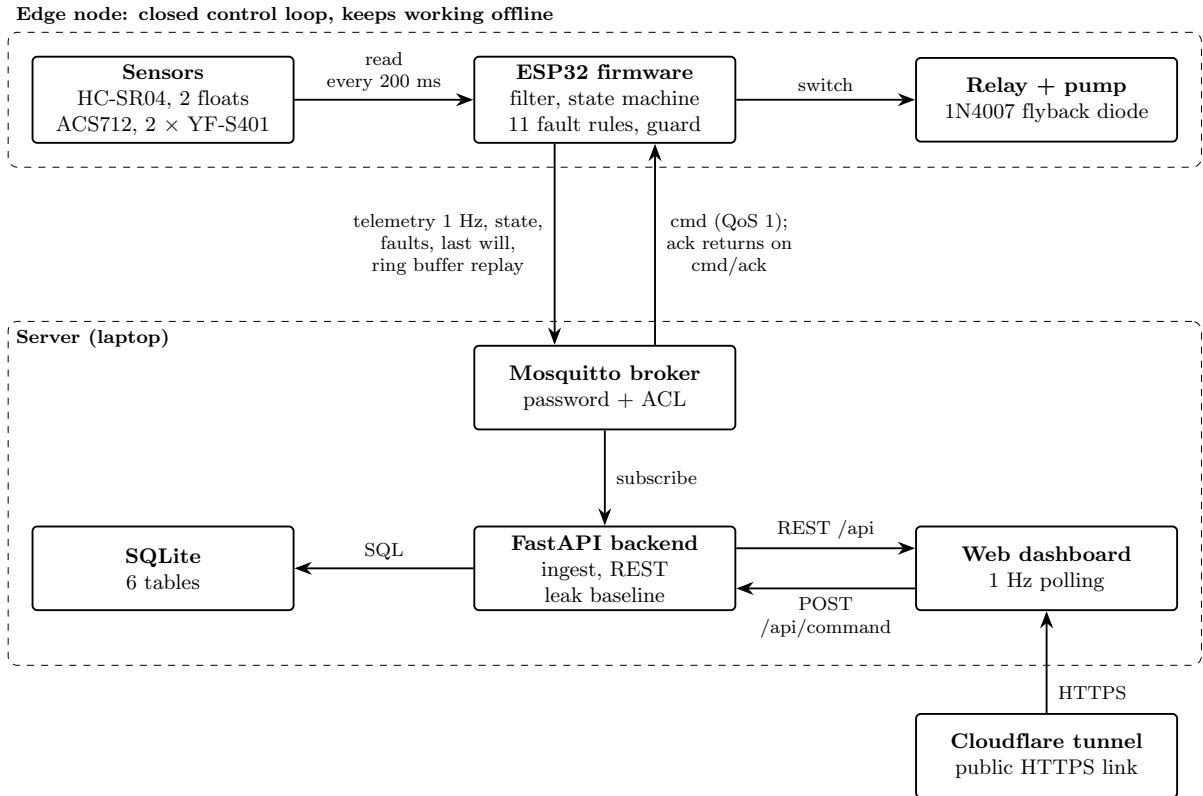


Figure 2.1.1: Architecture of the smart water tank system. The closed control loop lives entirely inside the edge node.

Mapping onto the five layer model of Section 1.4: the perception layer is the ESP32 with its sensors and relay; the transport layer is a Mosquitto broker over Wi-Fi; the processing layer is the ingestion process, SQLite and a FastAPI server; the application layer is the dashboard; and the business layer, kept deliberately minimal at this scale, is the daily volume report. The single most important property of Figure 2.1.1 is that the closed control loop lives entirely inside the first block, so the remaining blocks never sit on the path of a pump switching decision.

Wi-Fi was selected over BLE, Zigbee and LoRaWAN for three reasons. The system is mains powered, so the low energy advantage of BLE and LoRaWAN carries no weight. The device sits indoors within a few tens of metres, comfortably inside Wi-Fi coverage. Most importantly, Wi-Fi gives the node a native IP address so it speaks to the broker directly, whereas every alternative needs an edge gateway to translate between two protocol worlds [17], adding hardware and a point of failure. The conclusion reverses for tanks spread over a wide area or for a battery powered node. One practical constraint surfaced on the bench: the ESP32 radio supports only the 2.4 GHz band, so the node cannot join a 5 GHz only access point, and the firmware therefore accepts up to three network credentials and tries them in turn.

2.2 Function placement and edge autonomy

This is the central architectural decision of the project. The assignment requires that level control keep working during a network outage. Combined with the placement rule of Section 1.4, which states that any operation needing an immediate safe stop must be computed at the edge [1], the group adopted

a single principle: **the server never issues a direct pump switching command in automatic mode, and never bypasses the device’s own checks in manual mode.**

The edge node therefore owns sensor reading and filtering, the control state machine, every safety interlock, all fault detection rules, the pump health classifier and volume accumulation, because each of these demands a deterministic cycle and has to survive an outage. The backend owns only the things the node cannot do well: durable history, the daily volume report, and statistical baseline leak detection, which needs a window of several days and exceeds the memory available on the microcontroller.

This split makes the system degrade in a controlled way when connectivity is lost: the user loses remote visibility and remote commands, but the tank is still refilled at the correct thresholds and every safety constraint is still enforced. The behaviour is verified experimentally in Section 5.5.

2.3 Sensor selection and signal conditioning

Table 2.3.1: Main bill of materials.

Component	Role
ESP32 DevKit V1	Edge microcontroller
HC-SR04 ultrasonic sensor	Tank level, transducer face 15.88 cm above the tank floor
Two YF-S401 Hall turbine flow sensors	Inlet (pump to tank) and outlet (tank to consumer) flow
ACS712 Hall current sensor, 5 A variant	Pump current, used for the pump health classifier
Two mechanical float switches on the storage tank	Upper level overflow guard and lower level fill permission
Single channel relay module on GPIO 10, supplied from 3.3 V; 5 V submersible pump	Switching stage and actuator
470 μ F on the 5 V rail; 100 nF and 100 μ F at the ESP32 supply pin; 1N4007 and 100 nF at the pump	Bulk storage for motor inrush, local decoupling, inductive protection and brush noise filtering
Four 10 k Ω / 20 k Ω dividers; 220 Ω LED resistor	Bring the 5 V outputs of the echo, both flow sensors and the current sensor down to 3.3 V
12 V input on a screw terminal; two step-down modules, 12 to 5 V and 12 to 3.3 V	5 V rail: ESP32, ultrasonic, flow and current sensors; 3.3 V rail: relay module, floats, reset button

The tank. The storage tank has a square base of 10 by 10 centimetres, so one centimetre of level is exactly 0.1 litres. The transducer face sits 15.88 centimetres above the floor, measured with a ruler as six and a quarter inches. The working range is the lowest 12.4 centimetres, which the firmware treats as 100 percent, giving a usable volume of 1.24 litres and leaving 3.5 centimetres of air below the sensor at a full tank, outside its minimum range. The small volume is deliberate: it makes a full fill and drain cycle last a few minutes, so many control cycles can be observed in one bench session.

Level sensing. Non contact ultrasonic measurement was chosen because the transducer never touches the water, so it neither corrodes nor accumulates deposits, and its output does not depend on water conductivity. Two properties of this particular installation shaped the software in Chapter 3. First, the beam of an HC-SR04 spreads over roughly 15 degrees and is already about 8 centimetres wide when it reaches the water, against a 10 centimetre tank, so the edge of the beam strikes the wall on every ping and a fraction of echoes return from the wall or the floor rather than the surface. Second, the sensor occasionally locks up after a supply disturbance and stays silent until its supply is cycled, which no firmware reset can reach. The first property is handled by a robust filter; the second by a timeout rule and a documented recovery procedure. The calibration of Section 5.2 found a third property: when the water surface is 7.9 to 8.7 centimetres below the transducer, that is at a level of about 7.2 to 8.0 centimetres or 58 to 64 percent, the sensor does not return the surface at all but a stable stray echo, reading 1.0 centimetre or a distance beyond the floor in every one of forty pings. Because the stray value is steady it cannot be filtered out as noise; Section 3.2 explains how the firmware crosses this band on its model.

Echo line. The echo line was first routed through a TXS0108E automatic direction level translator. That part is designed for open drain buses and its internal pull ups and edge accelerators distorted the echo pulse, whose width is the only carrier of information. Replacing it with a plain 10 k Ω and 20 k Ω divider removed a 28 to 30 percent echo loss, as recorded in Appendix A.

Flow sensing. Both inlet and outlet carry a YF-S401 turbine sensor, where pulse frequency relates to flow as $Q = f/K$. Because the assignment makes the flow sensor mandatory, the group states its status precisely rather than simply as met: the hardware requirement is implemented, the measurement path is only partly validated, and control and accounting use flow derived from the level.

The history explains why. Until 22 September both signal lines carried about 1500 hertz of interference while the pump ran, forty times the real pulse rate, and a flyback diode and physical separation of the wiring did not reduce it (Section 5.2). The firmware was therefore given two sources of flow, selected at compile time: the pulse counters, or flow derived from the level rate described in Section 3.2. The prototype runs with the second. On 5 October the power stage was rewired: the relay now switches the supply ahead of the current sensor and the pump, the pump leads carry a diode and a capacitor at the motor, a 470 microfarad capacitor sits on the controller board, and the sensor lines were rerouted. The firmware also ignores any edge that arrives less than 1.2 milliseconds after the previous one, a pulse width filter that rejects spikes while passing the real pulses, which are about 30 milliseconds apart at these flows. After that change the inlet sensor reads 0 hertz with the pump off and 20 to 30 hertz with it running, which is water rather than interference. Each flow signal reaches its pin through a 10 k Ω

and 20 k Ω divider, the same as the echo line. Its factor was calibrated against the ruler and the tank geometry, $K = 85.5$ pulses per second per litre per minute instead of the nominal 98, and its reading is now published as `sens_in_lpm` and drives the level model of Section 3.2. The outlet sensor remains of limited use: the bench drain of 0.1 to 0.2 litres per minute is below the 0.3 litre per minute starting flow of the YF-S401, so it turns only intermittently.

Float switches. The two floats are not there to measure level. The upper float forms an overflow guard that operates independently of the ultrasonic path, and the lower float gives the pump a second, independent permission to start when the tank is low. Because two independent sources exist, the firmware can detect a contradiction between them and conclude that one reading is wrong. This is how the system satisfies the requirement to handle implausible sensor readings, as detailed in Section 3.4. On the board each float, like the fault reset button, switches its input to the 3.3 V rail (Figure 2.4.1). Float polarity was measured rather than assumed, because a float wired with the opposite sense reads as permanently engaged.

Current sensing. The ACS712 output passes through a 10 k Ω and 20 k Ω divider, which scales it by two thirds and moves its no load point from 2.5 volts to about 1.67 volts, inside the converter span. The 5 ampere variant produces 185 millivolts per ampere, which becomes 123 millivolts per ampere at the pin, or 8.11 milliamperes per millivolt. The firmware subtracts a startup offset averaged over 200 samples, reports the result in milliamperes, and reports zero below a 4 millivolt noise floor, since with the pump off the raw reading wanders between 0 and 2.7 millivolts, a single converter step being 0.8 millivolts. The measurement cannot resolve currents below about 30 milliamperes, which is enough to separate a running pump from a stopped one but not to characterise the motor, and Section 5.7 returns to this.

2.4 Power stage, supply and mechanical build

The pump is switched through a relay module. A power MOSFET was considered, since the lectures favour it in edge designs of modest power thanks to its very low on resistance [1], but the relay was chosen because a pump running in two states at a low switching rate never exercises the frequency advantage of a MOSFET, and because the module drives the load from its own supply. Inductive protection uses a 1N4007 diode antiparallel across the pump terminals, alongside a 100 nF ceramic capacitor at the motor terminals whose purpose is not spike suppression but damping of brush noise. The relay polarity was determined by measurement: the module fitted is active high, and an earlier configuration that assumed active low made every stop command start the pump, as recorded in Appendix A. The relay control pin is driven to the off level as the very first statement of the firmware, before the serial port or any sensor is initialised. The relay input moved to GPIO 10 on 5 October. That pin is wired to the flash chip on most ESP32 modules and is free only when the flash runs in dual line (DIO) mode, so the build sets `board_build.flash_mode = dio`; the group confirmed on the bench that the pin then switches the relay without resetting the chip.

Figure 2.4.1 is the schematic of the controller board. A 12 volt input on a screw terminal feeds two

step-down modules. The 5 volt rail supplies the ESP32 through its 5 V pin, the ultrasonic sensor, both flow sensors and the current sensor, with a 470 microfarad capacitor on the rail and a 100 nanofarad and a 100 microfarad capacitor at the ESP32 supply pin; the large capacitor buffers the inrush of the motor, whose start peak would otherwise pull the rail below the brownout threshold and reset the chip [1]. The separate 3.3 volt rail supplies the relay module, the two floats and the fault reset button. The pump circuit itself is wired off the board: the relay contact feeds the current sensor and then the pump, so the measured current is the pump current only. Figure 2.4.2 shows the assembled rig.

The rig forms a closed water loop: a source bucket, the pump, the inlet flow sensor, the transparent main tank, a manual valve acting as the consumption load, and the outlet flow sensor. The ultrasonic bracket must be rigid, because vibration makes readings oscillate and would be misdiagnosed as a sensor fault. The circuit was assembled in steps, each with its own test program in the repository (`test_level`, `test_relay`, `test_flow`, `test_current`, `test_floats` and others) that had to pass before moving on.

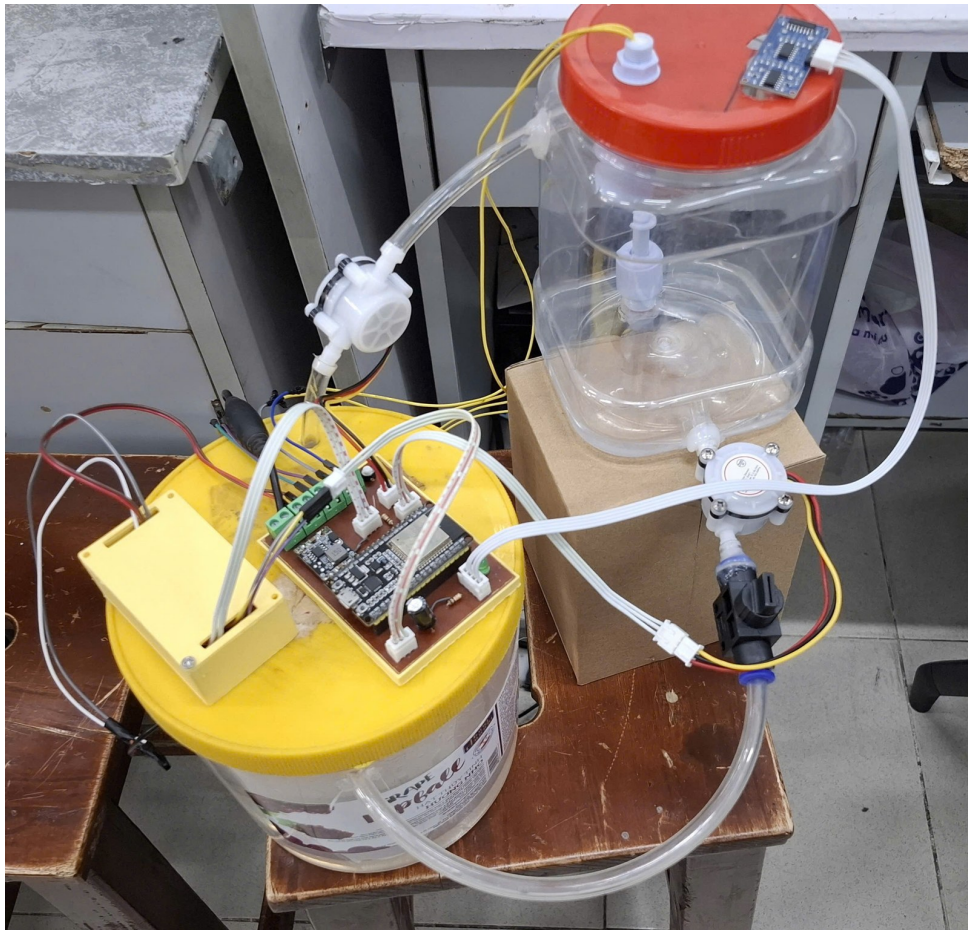


Figure 2.4.2: The assembled prototype: the main tank with the ultrasonic sensor and the upper float in the lid, the inlet and outlet flow sensors, the drain valve, and the controller board on the lid of the source bucket that holds the pump.

2.5 Design principles and topic tree

Three principles govern the protocol specification. First, every message carries a device generated timestamp, because although TCP preserves ordering on a single socket, brokers process multiple threads

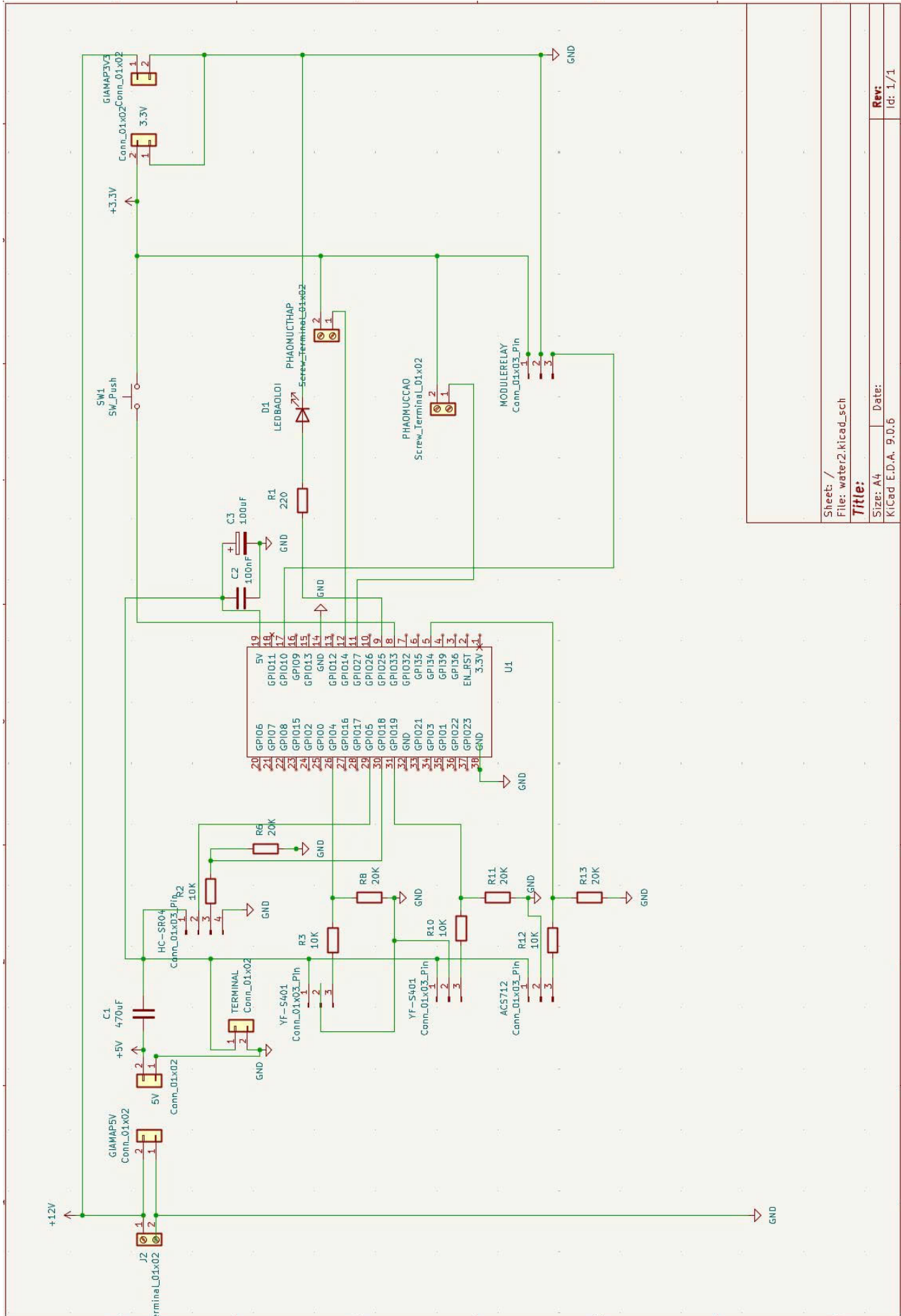


Figure 2.4.1: Schematic of the controller board (KiCad). Signals: TRIG GPIO 5, ECHO GPIO 18 through 10 k Ω /20 k Ω ; flow inlet GPIO 4 and outlet GPIO 19, current sensor GPIO 34, each through 10 k Ω /20 k Ω ; relay input GPIO 10; upper float GPIO 27, lower float GPIO 14 and reset button GPIO 33, each switched to 3.3 V; fault lamp GPIO 25 through 220 Ω .

concurrently so final arrival order may differ, and deterministic ordering must therefore be imposed at application level inside the payload itself [5]. Second, the quality of service level is chosen per topic according to the consequence of losing a message. Third, every downward command must be answered by an upward acknowledgement carrying the same identifier.

The topic root is built from broad scope down to the individual device as `wt/lab1/esp32-01`, following the hierarchical naming rule given in the lectures [5].

Table 2.5.1: MQTT topic tree and the reasoning behind each setting.

Topic	QoS	Retain	Reasoning
<code>.../telemetry</code>	0	No	One lost sample in a continuous 1 Hz stream changes no conclusion about the trend, and in exchange latency is lowest with no device buffer
<code>.../state/pump</code>	1	Yes	Published only on change, so a loss leaves the dashboard wrong until the next transition
<code>.../event/fault</code>	1	No	Losing an alarm is incorrect behaviour for a safety function
<code>.../status</code>	1	Yes	The last will payload is published once, with no retransmission
<code>.../cmd</code> and <code>.../cmd/ack</code>	1	No	A lost command or acknowledgement is a direct functional failure

The design respects all three naming pitfalls listed in the lectures [5]: no leading separator, no whitespace inside the path, and six explicit subscriptions rather than a root wildcard. No topic uses the highest quality of service level, because it requires a four step handshake and packet identifier tracking on the edge device, while the duplicate delivery problem it solves is already handled at application level: each command carries a unique identifier, so a duplicate simply causes the device to resend the same acknowledgement without acting twice. This is the idempotency property of Section 1.7.

2.6 Message structure and fault codes

Payloads use JSON built with a statically allocated document, following the rule that forbids dynamic allocation inside the loop [10, 11].

```
{ "dev": "esp32-01", "ts": 1790160600, "ts_ms": 1790160600123, "seq": 10421,
  "level_pct": 62.4, "level_cm": 7.74, "level_ok": true,
  "level_rate_cms": 0.036, "level_model_cm": 7.71, "blind_ms": 0,
  "flow_lpm": 0.36, "volume_l": 143.62, "volume_today_l": 12.40,
  "flow_out_lpm": 0.16, "volume_out_l": 98.10, "flow_src": "level",
  "sens_in_lpm": 0.35, "sens_out_lpm": 0.00, "sens_in_l": 21.80,
  "pump": true, "state": "FILLING", "mode": "AUTO",
  "current_mv": 36.1, "current_ma": 390.2, "pump_health": "ok",
  "float_max": false, "float_min": false, "fault": "", "rssi": -55,
```

```
"lv_try":605, "lv_ok":563, "rj_spread":31, "ctl_gap_ms":201 }
```

Listing 2.1: Telemetry message payload, abridged.

The `seq` field is a monotonically increasing counter allowing lost messages to be counted exactly in Section 5.5. The `ts_ms` field is produced by the device after time synchronisation and supports latency measurement. The `level_ok` field states whether the reading can be trusted, letting the interface distinguish a genuinely low tank from a broken sensor. The `flow_src` field states whether the flow figures come from the pulse sensors or are derived from the level, so no consumer can mistake one for the other. The `sens_*` fields carry the turbine readings alongside, so the two estimates can be compared in the stored data. The `lv_try`, `lv_ok` and `rj_*` counters record how many level pings were attempted, accepted and rejected at each filter stage; they turned out to be the most useful diagnostic in the project, as Appendix A shows. The `ctl_gap_ms` field reports the longest control period since the previous live message, so the first message after an outage proves whether the control loop was ever stalled by the network. Messages replayed after reconnection carry a `replay` flag and a reduced field set, and are excluded from latency statistics. The full message runs to several hundred bytes, so the serialisation buffer is 1024 bytes and the MQTT client buffer 1200 bytes; Appendix A explains why both margins are explicit.

Eleven fault codes are generated on the device and one on the backend. They fall into three groups. The self clearing group, `SENSOR_TIMEOUT`, `SENSOR_CONFLICT` and `LEVEL_LOST`, concerns signal quality and resolves on its own once the signal has been valid for long enough. The latching group, `OVERFLOW`, `NO_CURRENT`, `DRY_RUN`, `NO_PROGRESS`, `FILL_TIMEOUT`, `HARD_LIMIT` and `VOLUME_LIMIT`, requires the operator to confirm that the physical cause has been dealt with before the fault can be cleared. The warning group, `LEAK_SUSPECTED` on the device and `LEAK_BASELINE` on the backend, raises an alert without locking the pump. Every fault message also carries the pump health label and the pump current at the instant of the fault, captured before the pump is stopped, so the record states not only that something failed but which side of the pump failed. Trigger conditions are given in Section 3.4.

2.7 Command and acknowledgement mechanism

This section implements the feedback sync principle of Section 1.7 and satisfies the requirement that commands be acknowledged with a verified state.

```
// .../cmd
{"cmd_id":"a7f3c2e1","action":"pump","value":true,"ts":1735689600}
// .../cmd/ack
{"dev":"esp32-01","cmd_id":"a7f3c2e1","status":"rejected",
 "reason":"overflow_guard","state":"OVERFLOW_LOCK","pump":false}
```

Listing 2.2: Command message and matching acknowledgement.

The acknowledgement reports not only the outcome but also the actual device state after processing, so the dashboard always displays a state the device has confirmed rather than one the user has just requested. Four actions are supported: change mode, switch the pump, clear a fault and reset the volume counters. Ten rejection reasons are defined: `auto_mode_active`, `fault_active`, `overflow_guard`,

`float_max_triggered`, `water_too_close`, `level_sensor_invalid`, `min_off_time`, `no_active_fault`, `condition_still_present` and `unknown_action`. The dashboard translates each into a plain sentence, and for `min_off_time` it shows a countdown on the pump button until a start will be accepted.

The rejection table is worth more than a debugging aid: during the demonstration, pressing the pump on button while a fault is active and seeing an explicit rejection with a named reason is visible proof that manual mode cannot override the interlocks.

2.8 Connection lifecycle and availability monitoring

The firmware implements the four state connection machine described in the lectures. The critical point is the accompanying design note: connection logic must never sit inside a blocking wait loop, because that stalls the processor and suspends both analog sampling and safety checks for the whole outage [5]. Our reconnection routine therefore never blocks, retries only once its interval has elapsed, and the control loop runs independently of network state. The retry interval grows exponentially with an upper bound, $T_{retry} = \min(T_{max}, T_{base} \cdot 2^n)$ with T_{base} of 1 second and T_{max} of 30 seconds [18].

A non blocking design still has to account for the calls it cannot avoid. Opening a TCP connection to an unreachable broker blocks inside the network stack, and the default timeout of the ESP32 client is three seconds, during which the control loop does not run. The firmware therefore opens the socket itself with a 500 millisecond limit before handing it to the MQTT client, and bounds the wait for the broker's connection acknowledgement to two seconds. Two further bounds came from the outage test of Section 5.5. Writing to a socket whose peer has silently vanished blocks until the transmit buffer drains, which in the first firmware stalled the control loop for up to ten seconds; the firmware now checks, without waiting, whether the socket can accept data before every publish, and closes it once it has refused data for three seconds, so the ring buffer of Section 4.7 takes over after about three seconds instead of after the 15 second keep alive. And a new Wi-Fi association is attempted at most once every ten seconds, because restarting it every second, as an earlier version did, never left the radio enough time to finish joining. After a Wi-Fi reconnection the device also resolves the broker's host name over multicast DNS, because a laptop that moves between networks receives a different address on each; a fixed fallback address is used if the name does not resolve.

On connecting, the device pre registers a last will message announcing that it is offline, then overwrites it with an online message once the session is established. Should the node suffer an abrupt loss of power, the broker keep alive supervisor publishes the stored will by itself [5], so the interface switches to a disconnected state with no polling logic on the application side. A second backend check treats the device as offline when the newest telemetry message is older than 5 seconds. The two complement each other, since the will reacts quickly to a total power loss while the freshness check also catches a live session whose data stream has stopped.

Chapter 3

CONTROL AND SYSTEM INTELLIGENCE

3.1 Overview

This chapter presents the main technical contribution of the project. All of it runs in the microcontroller firmware, with the single exception of the rolling baseline leak detector of Section 3.6, which sits on the backend for the reason given in Section 2.2. The control loop runs on a fixed 200 millisecond cycle of six steps: read the sensors, filter the level, update the parallel level model, evaluate the fault rules, run the state machine, and drive the outputs. The loop does not depend on network state; the only network work inside it is bounded as described in Section 2.8.

3.2 Computed quantities

Level. The ultrasonic sensor measures the distance d from the transducer to the water, and the level is $h = 15.88 - d$ centimetres, then corrected as $h_{corr} = a \cdot h + b$ with coefficients from the calibration of Section 5.2. Because a share of echoes in this narrow tank return from the wall or the floor (Section 2.3), a single ping is never trusted. Each 200 millisecond cycle fires one ping and the reading passes through a chain of gates, each of which counts its own rejections so that the chain can be tuned from evidence rather than intuition:

1. **Window and geometry.** The last fifteen pings are kept; pings outside the physical geometry of the tank are discarded, in particular any distance more than 1 centimetre beyond the floor, which only a stray echo can produce.
2. **Spread.** If the interquartile range of the window exceeds 3.5 centimetres the window is rejected, because the sensor is then alternating between two surfaces and no single statistic of the window means anything. Appendix A explains why this is the interquartile range and not the full range.
3. **Lower quartile.** The 25th percentile of the window becomes the candidate reading. Like the median it tolerates isolated outliers that would drag a mean, and it adds a deliberate bias towards

the shorter distance: a weak echo is detected one or more 40 kilohertz cycles late, about 0.43 centimetres per cycle, so errors in this tank only ever lengthen the distance, and the shorter readings are the trustworthy ones.

4. **Rate.** The candidate is rejected if it differs from the last accepted value by more than physics allows in the elapsed time plus a 2 centimetre noise allowance, and, while the pump runs, if it lies more than 1 centimetre below the filtered level, since the tank cannot drain faster than a running pump fills it. An earlier version compared against the last accepted reading, and after one rejection that reference could never catch up, so the gate locked itself shut.
5. **Alpha-beta filter.** Accepted readings feed a two state filter that tracks both level $\hat{h}$ and its rate of change $\hat{v}$:

$$\hat{h}^- = \hat{h} + \hat{v} \Delta t, \quad r = h - \hat{h}^-, \quad \hat{h} \leftarrow \hat{h}^- + \alpha r, \quad \hat{v} \leftarrow \hat{v} + \frac{\beta}{\Delta t} r \quad (3.1)$$

with $\alpha = 0.03$ and $\beta = \alpha^2/(2 - \alpha)$. The earlier exponential moving average lagged a rising level by a constant offset; the alpha-beta filter tracks a steady ramp without lag, and its rate estimate $\hat{v}$ is used below for flow and for pump diagnosis.

If no reading is accepted for 5 seconds the level is marked untrusted (`level_ok` false). Independently of all of the above, a raw distance guard reads the unfiltered distance on every ping and declares the water too close after three consecutive readings nearer than 4.5 centimetres to the transducer. That guard shares nothing with the filter chain, so a filter defect cannot disable it.

Parallel level model. During a fill the sensor sometimes produces no accepted reading for tens of seconds, and the earliest firmware stopped every such fill at about half the target. The firmware now runs a model of the tank in parallel with the measurement, integrating $dh/dt = (Q_{in} - Q_{out})/A$ and pulling the model towards each accepted reading with a gain of 0.05. The inflow Q_{in} is the inlet turbine reading whenever it exceeds 0.05 litres per minute; otherwise it is the calibrated pump delivery while the pump runs and is not labelled `no_flow`, and zero when it is stopped. Q_{out} is the latest drain estimate. Feeding the model the measured inflow matters because the pump delivery changes with the level of the source bucket, from 0.36 to 0.24 litres per minute on different days, and a model that drifts cannot be used to judge whether a new reading is believable.

While the level is untrusted the state machine controls on the model. This is how a fill crosses the blind zone of Section 2.3, which takes 40 to 75 seconds at bench flow rates. The model is never allowed to replace the sensor for long: after 90 seconds without a single accepted reading the pump stops and `LEVEL_LOST` is raised, and a returning reading that disagrees with the model by more than 2.5 centimetres is rejected rather than adopted. A previous revision accepted such a reading after it had repeated for several cycles, on the argument that a value that keeps returning is real. The blind zone refuted that argument: its stray echo is perfectly repeatable, and the rule let it through as a level of 0 percent, which started the pump every 40 seconds. That acceptance path is now disabled. The overflow float, the raw distance guard and the run time limits keep working throughout any blind interval.

Flow. Section 2.3 explains why the turbine readings are not used in this installation. With flow derived from level, the reported inlet flow and the pumped volume use the calibrated pump delivery of

0.36 litres per minute while the pump runs and zero otherwise; the turbine reading is published beside it for comparison but does not yet feed the volume counter. Outlet flow is computed from the rate estimate as $Q_{out} = -\hat{v} \cdot A \cdot 60/1000$ litres per minute, but only when the pump has been off for at least 20 seconds and the surface has settled; the estimate is smoothed and then *held* while the pump runs. An earlier version computed outlet flow as pump delivery minus the net level rate at every instant. Because the delivery term switches instantly while the rate estimate lags, every pump start made the outlet curve jump by the full pump delivery and decay back, which on the dashboard looked as if consumption rose whenever the pump started. Holding the drain estimate removed the artefact and matches the physics: the valve setting, not the pump, determines the draw.

Volume. Accumulated volume is the numerical integral of flow,

$$V(t) = \sum_i Q_i \cdot \frac{\Delta t}{60}, \quad (3.2)$$

computed on the microcontroller rather than the server, because if it ran on the server every message lost during an outage would become water permanently unaccounted for. The totals are written to non volatile storage every 60 seconds, so a sudden power loss costs at most one minute. Section 5.2 measures the resulting volume error at -5 percent on average.

Current. Pump current is reported in milliamperes using the scale factor of Section 2.3, with readings under the 4 millivolt noise floor reported as zero.

3.3 State machine and hysteresis

Table 3.3.1: Controller state set.

State	Entry condition	Exit condition
BOOT	Power up	A valid level reading is available
IDLE	Pump off, no fault	Automatic mode and (level below 30 percent or lower float) and the safety guard allows a start
FILLING	Automatic start	Level above 70 percent after at least 3 s, or switch to manual
MANUAL_ON	Manual start accepted	Manual stop, return to automatic, or the safety guard objects
FAULT_SENSOR	Sensor fault raised	Valid, consistent signal for 2.5 continuous seconds
FAULT_DRYRUN	Pump fault raised	Operator clears the fault
OVERFLOW_LOCK	Overflow or run limit fault	Operator clears the fault and the overflow condition is gone

The reason for a state machine rather than a plain conditional expression is that fault states need to be sticky. A conditional rule would allow the pump to restart the moment a transient fault condition disappears, for instance when the suction tube touches the water again; the state machine holds the system in the fault state until an explicit intervention occurs. One invariant is enforced at the top of every cycle: whenever a fault code is set, the state must be a fault state. An early version let the manual and filling states return to idle without clearing the fault code, which left a device that looked idle and healthy while silently refusing every start.

The transition between `IDLE` and `FILLING` implements equation (1.2) with thresholds of 30 and 70 percent, leaving a 15 percent margin up to the 85 percent safety limit to absorb overshoot. Two timing constraints complement the band: a minimum on time of 3 seconds and a minimum off time of 20 seconds. Hysteresis rejects measurement noise near the threshold, while the timing constraints reject genuine short period oscillations and protect the pump from repeated inrush.

The minimum on time deserves a note, because the first draft set it to 10 seconds and that value was unsafe. At the measured 0.36 litres per minute a forced 10 second run adds 0.06 litres, nearly 5 percent of this 1.24 litre tank, while the state machine has no authority to stop it. Three seconds is long enough to stop the relay chattering and short enough that the band still governs. A hysteresis width of 40 percent may look wide for a control system, but the tank is a buffer: the goal is to guarantee water availability while limiting pump starts, not to hold a set point.

3.4 Fault detection rules

The system implements eleven rules where the assignment asks for at least one. They are evaluated every cycle in priority order and the first rule that fires ends the chain. Firing a rule stops the pump, lights the fault lamp, and publishes a fault message.

Table 3.4.1: Fault detection rules and trigger conditions.

Code	Trigger condition	Action
SENSOR_TIMEOUT	No accepted level reading for 45 s, or for 90 s while the level model is valid	Sensor fault, self clearing
SENSOR_CONFLICT	Upper float engaged while level reads below 70 percent, or lower float engaged while level reads above 60 percent	Sensor fault, self clearing
OVERFLOW	Upper float, or raw distance guard, or level at or above 85 percent	Lock
DRY_RUN	Pulse flow below 0.25 L/min for 6 s with pump on; active only with sensor flow	Lock
NO_CURRENT	Pump commanded on, no current for 2 s	Lock
LEAK_SUSPECTED	Pump off yet inlet flow above 0.20 L/min for 60 s	Warning only
FILL_TIMEOUT	One run longer than 240 s	Lock
HARD_LIMIT	One run longer than 300 s, unconditionally	Lock
VOLUME_LIMIT	More than 2.5 L pumped in one run	Lock
NO_PROGRESS	Pump on 90 s without the level rising 1.5 cm	Lock
LEVEL_LOST	Filling blind on the model for more than 90 s	Sensor fault, self clearing

`SENSOR_CONFLICT` is a direct application of the independent source consistency principle of Section 1.8 [14]. With only an ultrasonic sensor the system cannot tell a right reading from a wrong one as long as the value stays plausible; a mechanical float turns a wrong reading into a detectable contradiction.

The assignment’s example rule, pump on with near zero flow, appears twice. `DRY_RUN` implements it literally on the turbine signal, and is disabled when flow is derived from level, because inlet flow is then a constant that can never fall below any threshold: leaving the rule enabled would have been a guard that could not fire. `NO_PROGRESS` implements the same physical idea on a measured quantity: a pump that draws current while the level does not rise is not moving water. Its mark only ever moves down with noise and moves up only on real progress, so a single high outlier cannot hide a stalled fill. Appendix A records a bench episode in which the level reading was false for the whole of a fill and this rule, which never consults the validity flag, still stopped the pump.

The sensor timeout is long by design. The ultrasonic sensor in this tank drops out for tens of seconds during normal fills, and crossing the blind zone takes up to 75 seconds; the parallel model bridges those gaps, and a short timeout would stop every fill early, which the first version did at 4 seconds and the 45 second version still did inside the blind zone. The limit is therefore 90 seconds while the model is valid and 45 seconds otherwise. The price is that a disconnected sensor is reported after up to 90 seconds rather than 4, which is acceptable because every overflow layer that does not rely on the sensor remains

active meanwhile.

3.5 Manual mode and safety interlocks

The assignment requires that manual mode must not override safety constraints. The group meets this with an architectural decision inside the firmware: **every code path capable of starting the pump is forced through one single guard function.** The function returns nothing when starting is permitted and a reason code when it is blocked, checking in order: an active fault, the upper float, the raw distance guard, level at or above the safety limit, an untrusted level, and the minimum off time.

An earlier revision listed an empty source tank as a further condition. It was removed, and the removal is worth recording: the source vessel carries no instrument, and the condition was in fact reading the lower float of the *storage* tank under a wrong label. Blocking the pump on a quantity the system cannot observe is not a safety measure but a permanent refusal to operate, and dry running is covered by rules that rest on quantities the system does observe: absent current and absent level progress.

Because both the automatic state machine and user commands pass through the same guard, verifying the interlock means reviewing one function rather than auditing the codebase. When a command is refused, the acknowledgement carries the reason. Moreover, while in `MANUAL_ON`, the guard is re-evaluated on every cycle, so lifting the upper float stops a manually started pump within one 200 millisecond cycle, and the fault cannot be cleared while the float is still up: the clear command is answered with `condition_still_present`. One exception keeps manual and automatic filling consistent: a pump that is already running is not stopped merely because the level reading has just become untrusted, for instance in the blind zone of Section 2.3. It continues on the tank model for at most 90 seconds, with the overflow limit applied to the model level; a new start still requires a trusted reading.

3.6 Leak and anomalous consumption detection

This is one of the two advanced options in the assignment. It is built as two mechanisms at two places in the system.

The edge mechanism is the `LEAK_SUSPECTED` rule: once the pump is off, the supply line should carry no flow at all. It runs locally and therefore survives an outage, but it only sees the pipe section carrying the sensor and needs enough flow to turn the turbine, so with flow derived from level it is inactive.

The backend mechanism detects a sustained abnormal consumption level, the small slow shift discussed in Section 1.8 [15]. The day is divided into 48 slots of 30 minutes, and each slot keeps a baseline mean and variance of the volume used in it, updated by an exponentially weighted moving average:

$$\mu_k \leftarrow \mu_k + \alpha(V_k - \mu_k), \quad \sigma_k^2 \leftarrow (1 - \alpha) [\sigma_k^2 + \alpha(V_k - \mu_k)^2] \quad (3.3)$$

with $\alpha = 0.2$. Once a slot has been seen on at least three days, an alert `LEAK_BASELINE` is raised when $V_k > \mu_k + 3\sigma_k$ holds for two consecutive slots. Slicing the day reflects that domestic consumption varies strongly with the hour, and the two slot requirement is an accumulation step in the spirit of the

cumulative sum method [15], suppressing alarms from a single unusual draw.

Two safeguards were added after the detector learned from bad data. A slot in which the counter went backwards, because it was reset, or in which more water was recorded than the pump can deliver in 30 minutes, is skipped entirely rather than learned, since early noisy flow readings had accumulated thousands of phantom litres. And the baseline is rebuilt from the stored history at every backend start. The honest status is that after four bench days no slot has reached the three day minimum, so the detector is built and tested but has not yet raised a genuine alert.

3.7 Pump health classification

The second advanced option in the assignment is to use the consistency between pump current and flow to classify pump faults. The firmware assigns the pump one of five labels on every cycle:

- **idle**: the pump is off.
- **starting**: on for less than 2 seconds, or less than 25 seconds with no evidence yet.
- **ok**: current is present and the water is moving, meaning the level rate $\hat{v}$ exceeds 0.010 centimetres per second (or the turbine reads flow, when that source is selected).
- **no_current**: commanded on but no current, pointing at the electrical side: an open wire, a failed relay, a dead motor.
- **no_flow**: current present but no water moving after 25 seconds, pointing at the hydraulic side: a dry suction, a blocked or detached pipe.

The label is diagnostic, not protective: stopping the pump remains the job of **NO_CURRENT** and **NO_PROGRESS**. Its value is that the same symptom, a tank that does not fill, is split into two causes that send the operator to two different places, and that the label is captured at the instant of every fault, before the pump is stopped, and stored with the fault record. The 25 second evidence window is chosen from the measured fill rate: at 0.036 centimetres per second of net rise, 25 seconds produce 0.9 centimetres, well clear of the filter noise.

Chapter 4

BACKEND, INTERFACE, SECURITY AND RELIABILITY

4.1 Backend architecture

The backend is a single Python process containing three concurrent components. The ingestion component holds a persistent broker session and subscribes to the six specified topics. The API server exposes endpoints for the dashboard and for extracting experimental data. The rolling baseline leak detector is invoked on every incoming telemetry message and is described in Section 3.6; because it learns slowly, it is rebuilt from the stored history each time the process starts, so a restart does not discard days of learning. In addition the process keeps an in memory copy of the latest state, so that the current state endpoint can answer without touching the database, since the dashboard calls it once per second.

For every telemetry message the ingestion component records the device timestamp alongside `recv_ts`, written by the backend on arrival, so that a one way delay $\tau = t_{recv} - t_{sample}$ is available from a single query without external instrumentation. The device timestamp is stored twice: `ts` at one second resolution, kept so that existing consumers of the payload continue to parse, and `ts_ms` at millisecond resolution, which is the field the latency query prefers. The coarser field alone carries a quantisation error of up to 500 milliseconds, larger than the transport delay it is meant to measure, and Section 5.5 records the corrected figures side by side. The measurement assumes both clocks are synchronised; messages emitted before synchronisation completes carry a zero timestamp and are excluded from the statistics. Because that assumption cannot be discharged on a microcontroller disciplined only by the network time protocol, the one way figure is treated as a comparison quantity and the command round trip, timed entirely by the server clock, is the reported result.

4.2 Data schema and API

The schema has six tables, split according to data behaviour, and is shown with the whole repository in the code guide delivered alongside this report: `telemetry` is written continuously at one row per

second, while `faults`, `pump_events`, `commands`, `availability` and `config` are written sparsely and discretely. The telemetry table is indexed on the timestamp column, the only column ever used as a filter predicate. The group applies the tag and field separation principle of Section 1.7: the device identifier acts as an indexed tag while the measured quantities are unindexed fields. In particular the sequence number column, although it changes on every row, is deliberately left unindexed to avoid exactly the cardinality explosion the lectures warn about [8]. The command table stores both the send time and the acknowledgement time, whose difference is the command round trip latency.

On database choice, the group is aware that the theory in Section 1.7 identifies real limits of relational engines for high rate time series data [8]. Those limits, however, appear at thousands of writes per second, whereas this system writes one row per second from one node, three to four orders of magnitude below the problematic regime. Deploying a dedicated time series engine would add operational complexity without any measurable benefit, so the group chose SQLite while still respecting the schema design principles above [19].

Table 4.2.1: Main API endpoints.

Method	Path	Purpose
GET	<code>/api/latest</code>	Latest state, data freshness, pending commands
GET	<code>/api/telemetry</code>	Time series over a requested interval
GET	<code>/api/volume/daily</code>	Volume pumped in and drawn out per day
GET	<code>/api/leak/baseline</code>	State of the rolling baseline, slots learned
POST	<code>/api/login</code> , <code>/api/logout</code>	Open or close an administrator session
GET	<code>/api/auth</code>	Whether this browser may control the pump
GET	<code>/api/faults</code> , <code>/api/events</code> , <code>/api/commands</code>	Fault log, pump switching log, command log with acknowledgement latency
POST	<code>/api/command</code>	Issue a command, returns its identifier; session required
GET, PUT	<code>/api/config</code>	Read thresholds; update requires a session
GET	<code>/api/stats/latency</code>	Latency statistics used in Section 5.5

Method selection follows the two properties introduced in Section 1.7. Read endpoints use GET because they are safe and change no state. The configuration endpoint uses PUT because replacing the full threshold set is idempotent. The command endpoint uses POST because each call creates a new command record with its own identifier and is therefore not idempotent [10].

Daily volume. The first implementation computed each day’s volume as the maximum minus the minimum of the accumulated counter. That is wrong whenever the counter is reset during the day, and it was: the result swung by litres after every reset. The corrected query uses a window function to take the

increment between consecutive rows and sums only increments that are positive and physically possible in one second, so a reset contributes nothing instead of a large negative step. Inflow increments are additionally counted only while the previous row showed the pump running, because with flow derived from level the inlet cannot be non zero with the pump off. Rows stamped before the device clock synchronised carry a timestamp near zero and are assigned to the server's arrival date instead, which removed a spurious day in 1970 from the chart. Figure 4.2.1 shows the result for the four bench days.

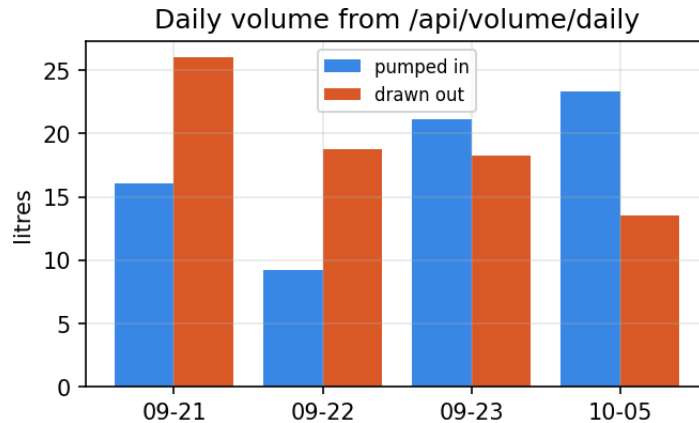


Figure 4.2.1: Daily volume pumped in and drawn out, as served by the API.

4.3 Monitoring interface

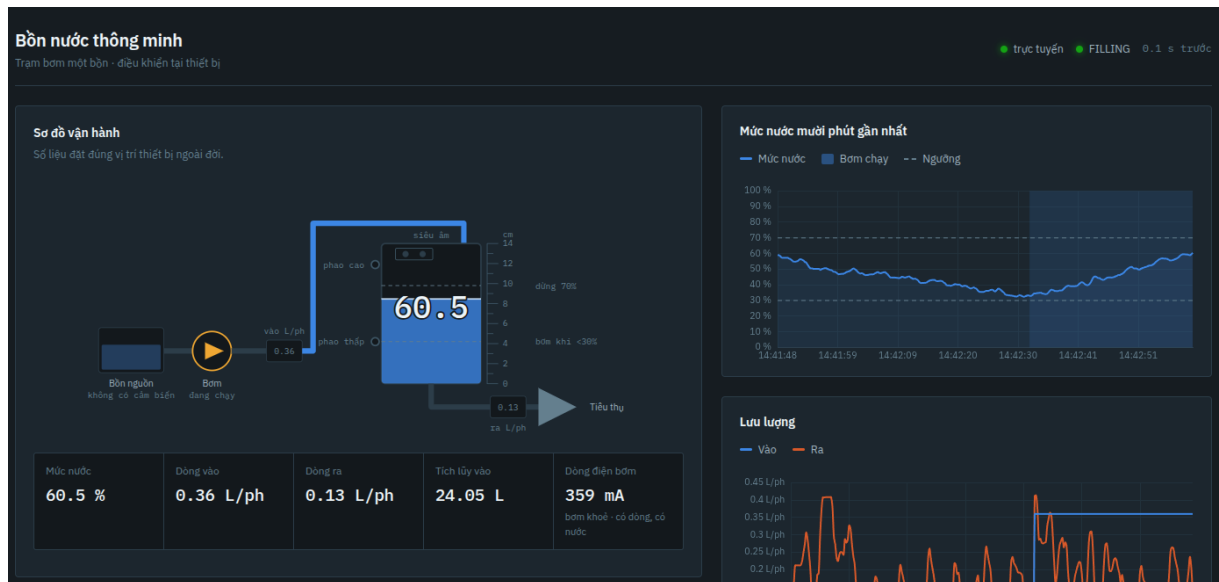


Figure 4.3.1: Monitoring dashboard during an automatic fill: process mimic with the tank at 60.5 percent, numeric tiles with the pump health label, and the level history. For this screenshot, recorded bench data from 23 September were replayed into an isolated copy of the backend, so the live database was not modified.

The interface is designed for an operator who glances at the screen for a few seconds to answer three questions: what is the level, is the pump running, and is anything wrong. It is a single HTML file with

no build step and no framework. The upper part is a process mimic in the style of a control room: the source bucket, the pump, the pipe and the tank are drawn as a diagram, the water in the tank rises and falls with the measured level against a ruler graduated from 0 to 12.4 centimetres, and the low, high and overflow thresholds are drawn at their true positions for the active configuration. Drawing the thresholds on the tank lets a viewer see at once where the level sits relative to the hysteresis band and why the pump just switched.

Below the mimic are the numeric tiles: level, inlet and outlet flow, pump current in milliamperes with the pump health label underneath it, and today's volume. Beside them are the history charts, a daily volume bar chart, and three logs: faults (with the pump health at the moment of each fault), pump events and commands. A freshness line next to the connection lamp shows how many seconds ago the last message arrived.

Two display details matter more than they appear to. The device reports once per second, so a number written straight into the page moves in visible steps; the interface eases the displayed value towards each new reading. And the filtered level still moves by a few millimetres from sample to sample, which on a large readout looks like instability; the display applies a deadband of 0.25 centimetres, below the resolution anyone can act on, so the number holds still unless the water really moves. Neither detail touches the data: the database and the API carry the unmodified values.

On the update mechanism, the lectures recommend REST for low frequency administrative tasks and WebSocket for live streams [18, 20]. The group chose polling at 1 Hz, which runs against that general recommendation and therefore needs justification. The device publishes at 1 Hz, so the polling rate equals the data production rate and no request is wasted. With a handful of clients and one node the resulting traffic is a few kilobytes per second. A WebSocket introduces a stateful channel with its own lifecycle to manage, including heartbeats and reconnection [18], that is a new set of failure states in exchange for a benefit that is not measurable at this scale. The conclusion reverses beyond roughly ten nodes or a required update rate of 5 Hz.

The feedback sync principle is implemented in how command state is displayed. When the user presses the pump button the interface does not change the pump symbol; it posts the request, receives an identifier, shows “sending command” and polls the command log for that identifier. When the acknowledgement arrives it reports the outcome in a plain sentence, including the reason for a rejection, while the pump symbol follows only the pump state reported in telemetry. An early version gave no feedback at all on a rejection, and a user reported that manual mode “did not work” when the device had in fact correctly refused a start inside the 20 second minimum off time; the countdown shown on the button now makes that interlock visible instead of mysterious.

Remote access and control rights. The assignment frames the system as remote supervision, so the dashboard is published through an outbound Cloudflare tunnel that gives it a public HTTPS address without opening any inbound port on the laboratory network. Anyone with the address can watch. Control is a separate matter: the control panel stays hidden until the browser holds an administrator session, obtained by entering a password kept in a file outside the repository. The group first tried to grant control by client address, allowing only the host machine, and abandoned it for two reasons found

in testing: through the tunnel every request arrives from the loopback address, so the rule granted control to the whole Internet; and on the host itself every open tab could control the pump, which is not what the owner meant by “only my machine”. Section 4.6 describes the session mechanism.

The interface adapts to the state reported by the device, following the conditional rendering principle [18]: data older than 5 seconds or an arriving last will message turns the connection lamp red; a fault or overflow lock switches the status to an alarm style; and a false `level_ok` flag marks the level as untrusted. Hiding or disabling buttons is only a convenience, since the backend rejects commands without a session and the device rejects unsafe ones regardless of who sent them.

4.4 Device simulator

The simulator implements the hydraulic model introduced in Section 1.8 together with a sensor noise model, and publishes messages that conform to the specification of Sections 2.5 to 2.8. It serves two purposes: while components had not arrived it allowed the backend and interface branches to be developed fully, and it allows control parameters to be swept across dozens of configurations in a few minutes, which is not feasible on real hardware. The sweep results are reported in Section 5.3. The simulator can also inject three failure modes, namely a pump that runs without delivering water, a sensor stuck at a fixed value, and a leak at a prescribed flow rate.

4.5 Threat model and mitigations

Under the ITU-T Y.2060 model of Section 1.4, security is a cross cutting capability acting on every layer rather than a layer of its own [2], so the analysis is organised per layer.

Table 4.5.1: Threat scenarios and mitigations.

Attack scenario	Consequence	Mitigation
Connect to the broker and publish a pump on command	Overflow, pump damage	Mandatory authentication, per topic access control
Sniff traffic to harvest credentials	Full control takeover	Not mitigated in the lab build: broker on plain port 1883, reachable only inside the laboratory network; TLS on port 8883 is the deployment step
Forge telemetry so the system misreads the level	Incorrect pumping, possible overflow	Local interlock based on the mechanical float
Replay commands repeatedly to make the pump chatter	Mechanical pump damage	Minimum on and off times enforced on the device
Call the API directly, bypassing the web interface	Unauthorised commands	Every control endpoint checks an administrator session
Guess the administrator password through the public address	Control takeover	Five failures in five minutes lock the source address

The second row is the one gap the group leaves open, and Section 4.6 states it without qualification. The two rows after it are worth highlighting because they are mitigated not by cryptography but by the control design itself. The mechanical float is a physical constraint that no network message can override, and the minimum switching times are enforced in firmware, so an attacker may send any number of commands without making the pump toggle faster than the limit. This illustrates the principle that safety in a cyber physical system cannot be separated from its control design.

4.6 Authentication, authorisation and encryption

The Mosquitto configuration disables anonymous connections entirely and mandates both a password file and an access control list file [21]. The group applies least privilege: the edge device may only publish on the northbound topics under its own branch and may only subscribe to its own command topic, while the backend may publish on the command topic and read the whole northbound branch. The practical meaning is that if the edge device is compromised, the attacker still cannot use that device to issue control commands, because the device account has no write permission on the command topic. This is exactly the node compromise scenario the lectures raise when introducing access control lists [5].

On the API side the system distinguishes a viewer, who needs nothing and may call every read endpoint, from an administrator, who may also call the command and configuration endpoints. Ad-

administrator rights are obtained by password. The password lives in a file outside the repository with owner only permissions, and is compared in constant time. A correct password creates a random 256 bit session token, returned as a cookie marked `HttpOnly` so page scripts cannot read it, `SameSite Strict` so another site cannot make the browser send it, and `Secure` whenever the request arrived over HTTPS, which through the tunnel is always. Sessions last seven days and are held in memory, so restarting the backend logs everyone out, which the group accepts at this scale. A missing or expired session returns 401. Five wrong passwords within five minutes lock further attempts from that address for the rest of the window, with the client address taken from the tunnel's forwarded header only when the connection itself comes from the local machine, since any remote caller could forge that header.

The group records one design it rejected. Control was first granted to requests from the host machine's own address. Behind the tunnel every request, from anywhere, arrives from the loopback address, so the rule silently gave control to the whole Internet; and even without the tunnel it gave control to every browser tab on the host rather than to a person. Identity has to come from something the person knows, not from where the packet appears to come from.

What is not encrypted. The system as built does not use TLS on the MQTT path, and the group does not claim end to end encryption. The broker listens on the unencrypted port 1883, inside the laboratory network only; the Cloudflare tunnel carries the dashboard and the API over HTTPS and never exposes the broker. Anyone able to capture Wi-Fi traffic on that network could therefore read the telemetry and the broker passwords of the device and the backend, which is exactly the sniffing threat of Table 4.5.1. Within the lab, the access control list limits what a stolen device password allows, and the control design limits what a forged command can do; neither replaces encryption. For a real deployment the group would move to MQTTS on port 8883 with a certificate on the broker [5, 7]. The cost should be stated plainly: the TLS handshake requires the microcontroller to perform public key operations, which noticeably increases startup time and memory consumption. One principle holds in every configuration, namely that no secret ever appears inside the telemetry payload.

4.7 Reliability during a network outage

This is a mandatory requirement and also the most important property of the system. Four mechanisms are implemented.

First, the control loop runs on a fixed cycle, and the network calls it cannot avoid are bounded: opening a socket to the broker gives up after 500 milliseconds instead of the default three seconds, waiting for the broker's acknowledgement gives up after two seconds, and no message is written to a socket that cannot accept it at once (Section 2.8). Measured on the final firmware with the link cut for 120 seconds, the longest control period during the outage was 0.88 seconds, against up to ten seconds before the socket check was added. During an outage the system therefore loses only remote visibility while retaining every control and safety function, and the longest control period is published in the first message after reconnection so the claim can be checked rather than assumed.

Second, when the link is down telemetry messages are not discarded but packed into a compact

structure and pushed into a circular buffer holding 240 entries, about four minutes of data. The circular buffer was chosen for exactly the reason given in the lectures, namely constant time insertion with no array shifting [1]. Each entry stores only the essential fields as scaled integers instead of a full JSON string. Once the link returns the whole buffer is replayed in order, tagged so it can be excluded from latency statistics. The buffer only starts filling once the firmware knows the link is down, which is why the socket is closed after three seconds of refused writes rather than left to the 15 second keep alive.

Third, accumulated volume is written to non volatile storage and reloaded at startup, so a sudden power loss does not erase the consumption history.

Fourth, inside the initialisation routine and before any sensor is read, the relay control pin is driven to the pump off level, so every restart begins from a safe state. The state machine then stays in `BOOT` until a first level reading has been accepted, so the zero that the level variable holds at power up can never be read as an empty tank (Section 5.3). The integrated brownout detector acts as an additional protective layer: when the supply falls below the critical threshold, flash writes become unreliable and the core may execute corrupted instructions, so the hardware deliberately holds the device in a safe reset state until the voltage recovers [1].

Table 4.7.1: Independent layers of overflow protection.

Mechanism	Location	Still effective when
High threshold of the hysteresis band	State machine	Sensor is healthy
Safety limit at 85 percent	Fault rule	Sensor is healthy
Raw distance closer than 4.5 cm, three pings in a row	Fault rule, unfiltered	The filter chain is producing nonsense
No level progress in 90 seconds	Fault rule	Level reading is frozen or false
2.5 litres in one run; 240 s and unconditional 300 s run limits	Fault rules	Every level sensor has failed
Upper mechanical float	Digital input	Ultrasonic sensor has failed
Relay driven off as the first instruction at boot	Firmware start up	Controller has just reset

The value of this arrangement lies in the independence of the layers. The first two both rely on the filtered ultrasonic level and fail together if it fails, but the raw distance guard bypasses the filter, the run limits consult only the clock, and the float requires no measurement at all. The time limits are honestly the weakest layer: a 240 second run at the measured delivery adds 1.44 litres, more than the whole 1.24 litre tank, so on their own they bound the damage rather than prevent it, and they sit behind the float rather than replacing it. No single point of failure disables the overflow protection as a whole.

Chapter 5

EXPERIMENTS, RESULTS AND CONCLUSION

5.1 Principles and reference instruments

Three principles apply to the whole experimental programme: every measurement must have a reference independent of the system under evaluation; every experiment must be repeated enough times to yield a dispersion figure and not merely a mean; and results must be reported through standard statistical indicators rather than qualitative remarks [22, 23]. The references are a millimetre scale standing in the tank, the geometry of the tank itself (a 10 by 10 centimetre base, so one centimetre of level is 0.1 litres), a measuring jug, and the server clock. The indicators are the per point error $e_i = x_{meas,i} - x_{ref,i}$, the mean absolute error and the root mean square error.

All figures in this chapter come from the database of the running system or from two scripts in the repository that read it: `tools/analyze_experiments.py` for the automatic cycles, switching and latency, and `tools/bench.py` for the interactive tests (level calibration, fault injection with a timestamped mark, and network outage). Two sessions are analysed. The control session is 23 September, 16:48 to 18:12, 4870 telemetry messages with 99.5 percent carrying a trusted level and no fault raised. The calibration and acceptance session is 5 October, after the power stage was rewired and the sensor geometry measured again; it ran the final firmware and produced the level calibration, the outage test with the pump running and the automated system test `tools/systest.py`. The test procedures are written out step by step in `docs/TESTING.md` so that any member can repeat them.

Table 5.1.1: Experiment plan and status.

ID	Subject	Reference	Status
E1	Level calibration and repeatability	Ruler, pumped volume, still water, lower float	Done, 12 points on 5 Oct
E2	Flow and volume error	Tank geometry, measured drain, ruler	Done, 6 cycles; inlet sensor calibrated
E3	Control response	Configured thresholds	Done, 6 cycles plus final firmware
E4	Switching frequency	Pump event log	Done, 84 min
E5	Fault detection and response	Fault and pump event logs	Partly done; NO_CURRENT not injected
E6	Network loss and recovery	Sequence numbers	Done on final firmware, pump running
E7	Telemetry and control latency	Server clock	Done, $n = 40$ on final firmware
E8	Control policy comparison	Simulator	Simulated
E9	Wi-Fi power saving	Server clock	Done

5.2 Sensor calibration

E1, level against a physical reference. The ruler run planned with `bench.py level` was replaced by a stepwise calibration that needs only one ruler reading, run on 5 October with `test_stepcal`. With the drain valve closed the pump runs in 20 second steps; after each step it stops, the surface is left to settle for 12 seconds, and forty raw pings are recorded. The inlet turbine counts every litre pumped, so the reference level after each step is the ruler reading plus the pumped volume divided by the 100 square centimetre base. The run is anchored to a single ruler reading of 4.9 centimetres at the end of the first pass, and the turbine factor was set from that same pass ($K = 85.5$, the nominal 98 counted 14.6 percent short), so the six low points test linearity rather than an independent scale. The second pass used the same factor without adjustment. The reading compared is the 25th percentile of the forty pings, the statistic the firmware uses.

Table 5.2.1: E1 level calibration, 5 October. Reference: ruler plus pumped volume over the tank base. Bold: blind zone.

Pass	Step	Reference (cm)	Ultrasonic (cm)	Error (cm)
Low, from a dry floor	1	0.92	0.72	−0.20
	2	1.73	2.31	+0.58
	3	2.54	1.75	−0.79
	4	3.33	3.81	+0.48
	5	4.11	3.72	−0.39
	6	4.90	4.92	+0.02
High, from the ruler reading	1	5.67	4.84	−0.83
	2	6.45	6.05	−0.40
	3	7.21	1.01	−6.20
	4	7.97	−5.42	−13.39
	5	8.71	7.17	−1.54
	6	9.48	9.57	+0.09

Outside the blind zone the ten points give a mean absolute error of 0.53 centimetres and a root mean square error of 0.68 centimetres, with the largest error, −1.54 centimetres, at the edge of the zone. The six low points alone give 0.48 centimetres. Inside the zone the sensor returns a stable stray echo: a level of 1.01 centimetres for a true 7.21, and a distance 21.3 centimetres away, beyond the floor, for a true 7.97. Figure 5.2.1 shows both regions. The geometry gate rejects the second value outright and the model gate of Section 3.2 rejects the first, so the firmware crosses the zone on its model; a stilling tube, foam around the transducer or a higher bracket would remove the cause.

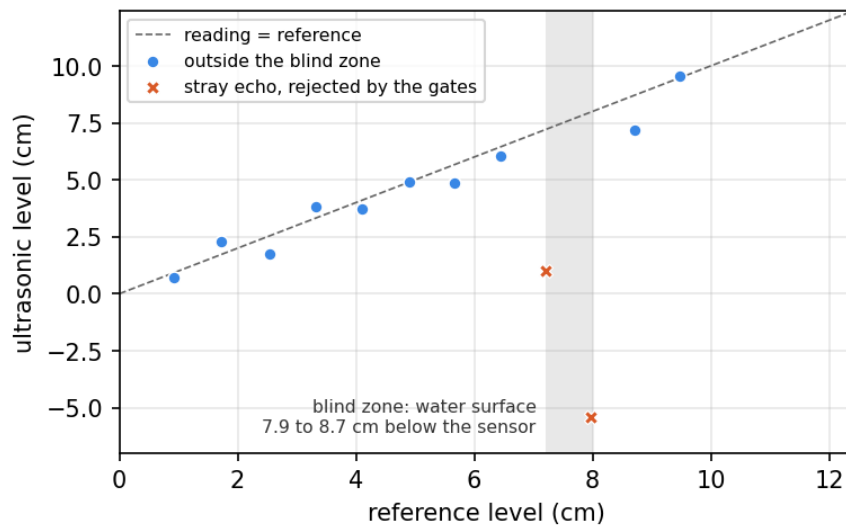


Figure 5.2.1: E1: ultrasonic level against the reference level. Crosses mark the blind zone, where the sensor returns a stray echo instead of the surface.

E1, repeatability. The database of 23 September adds two fixed references that need no ruler. Still water read twenty times in a row gives the noise floor. The lower float switch is a mechanical contact at a fixed height on the tank wall; every time it closed or opened, the level reported in the same second was recorded, so a level measurement that drifted or jumped would see the float at a different level each time.

Table 5.2.2: E1 level repeatability against fixed physical references, 23 September.

Reference	Mean level	n	Spread
Still water, consecutive readings	3.88 cm	20	sd 0.010 cm, range 3.87 to 3.89
Lower float closes	4.55 cm	8	sd 0.074 cm, range 4.46 to 4.67
Lower float opens	4.87 cm	9	sd 0.158 cm, range 4.63 to 5.11

The float closed at a reported 4.55 centimetres on average with a standard deviation of 0.074 centimetres over eight events; the 0.32 centimetre difference to the opening level is the float’s own mechanical hysteresis, not a measurement error. Both spreads are two orders of magnitude smaller than the width of the control band, so measurement noise cannot by itself make the pump switch. These figures were recorded with the sensor at its earlier height of 17.5 centimetres, before the bracket was refitted; they establish repeatability, while Table 5.2.1 establishes the absolute scale on the final geometry.

E2, pump delivery and flow sensors. The pump delivery was calibrated from the level: with the outlet closed the level rose 1.2 centimetres in 20 seconds, that is 0.12 litres in 20 seconds or 0.36 litres per minute, against 1.67 litres per minute in the datasheet at zero head. Until the rewiring of 5 October the turbine sensors could not be calibrated in the usual way, by counting pulses against a collected volume, because they did not count water. Table 5.2.3 shows why, and what changed.

Table 5.2.3: YF-S401 pulse rates on the bench, 22 September and 5 October.

Condition	Inlet	Outlet
Pump off, internal pull ups only	0.26 to 0.51 L/min apparent flow; exactly 50.00 Hz on one channel	
Pump off, 4.7 k Ω external pull ups	0 Hz	0 Hz
Pump running, before flyback diode	1474 Hz	1548 Hz
Pump running, diode fitted and wiring separated	1487 Hz	1567 Hz
Pulse rate expected for the real 0.36 L/min	about 35 Hz	
After rewiring, 5 Oct, pump off	0 Hz	0 Hz
After rewiring, 5 Oct, pump running	20 to 30 Hz	intermittent

The pull ups removed mains pickup completely. The pump interference, forty times the real signal, was unaffected by the diode and by moving the wiring apart, which places it in the shared supply and ground rather than in radiation. The outlet draw of 0.11 to 0.20 litres per minute is in any case below the 0.3 litre per minute starting flow of the sensor. After the rewiring of 5 October, described in Section 2.3, the inlet channel counts water: no pulses with the pump off, including with the drain valve closed and the pump stopped for 80 seconds, which also showed that no water flows back through the pump, and 20 to 30 hertz with it running. With $K = 85.5$ that is 0.24 to 0.35 litres per minute, the spread following the level of the source bucket. The calibrated inlet sensor now feeds the level model, but the volume counter and the control rules still use flow derived from the level (Section 3.2), because the turbine has been validated on one calibration run, not over complete cycles. The volume error below is therefore the error of the level derived estimate.

E2, volume estimation error. For each automatic fill the device’s pumped volume is compared with a reference built only from the level and the tank geometry:

$$V_{ref} = A \Delta h + \bar{Q}_{drain} T_{fill}, \quad (5.1)$$

where Δh is the level rise over the fill, T_{fill} its duration, and $\bar{Q}_{drain}$ the drain rate measured independently by fitting the level slope with the pump off for 90 seconds before the fill and 90 seconds after it (skipping 20 seconds after the stop while the pipe empties). Averaging before and after matters because the drain follows Torricelli’s law and is faster when the tank is fuller. The device value is the constant delivery integrated over the run, so the comparison tests the calibrated constant and the timing, not a sensor.

Table 5.2.4: E2 and E3: complete automatic fill cycles, 23 September.

Start	Stop (%)	Peak (%)	Over-shoot	Fill (s)	Drain (L/min)	V_{dev} (L)	V_{ref} (L)	Error
16:56:37	70.5	71.9	+1.47	237.4	0.180	1.289	1.520	−15.2%
17:09:56	70.8	72.4	+1.61	144.0	0.189	0.859	0.983	−12.6%
17:17:13	70.4	71.6	+1.20	146.0	0.146	0.872	0.875	−0.3%
17:32:34	70.3	71.5	+1.23	219.0	0.113	1.311	1.212	+8.2%
17:55:56	70.1	71.9	+1.85	144.0	0.196	0.864	1.006	−14.1%
18:02:00	70.4	71.5	+1.10	160.0	0.150	0.958	0.930	+3.0%
Mean	70.4		+1.41	175.1	0.16			−5.2%

The mean volume error is −5.2 percent with a spread from −15.2 to +8.2 percent. The sign pattern is informative: the three largest negative errors coincide with the three largest drain rates, which is what a constant delivery predicts, since a fuller source bucket and a shorter delivery head both change the real pump output while the model keeps 0.36. The error is therefore a property of treating delivery as a constant. The 5 October measurements confirm it: with a lower source bucket the same pump delivered 0.24 litres per minute, a third less than the constant. Moving the volume counter to the now calibrated inlet sensor is the remedy, and the next step.

5.3 Control response and switching

E3, control response. Figure 5.3.1 shows the analysed session. Every automatic fill stopped between 70.1 and 70.8 percent against a 70 percent threshold, and the level peaked 1.1 to 1.85 percent higher, the water still in the pipe after the relay opens. The worst peak, 72.4 percent, leaves 12.6 percent of margin to the 85 percent overflow limit. A fill from the lower threshold took 144 to 160 seconds at a net rise of 0.033 to 0.037 centimetres per second.

On the final firmware of 5 October the complete automatic fill of the outage test (Figure 5.5.2) stopped at 70.7 percent and peaked at 76.2 percent. The larger overshoot has a clear cause: the fill had just crossed the blind zone on the model, and the filter was still catching up with the surface when the threshold was passed. Two other automatic fills that evening stopped at 48 percent with `NO_PROGRESS` and the health label `no_flow`: with the drain valve as it was set then, the outflow at that head matched the pump delivery, the level hovered between 46 and 51 percent for 90 seconds, and the rule did what it is for, stopping a pump that was not gaining on the tank.

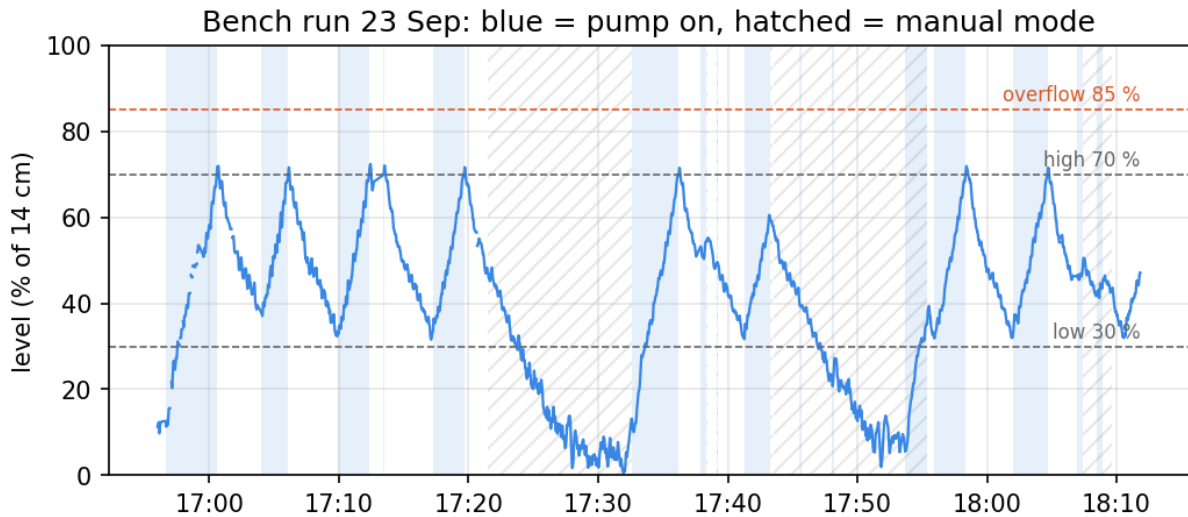


Figure 5.3.1: Level during the analysed session. Blue bands: pump running. Hatched: manual mode, used for command tests and for draining the tank below the band on purpose.

One result was not predicted. The automatic fills did not start at 30 percent but at 32 to 33 percent, and in every case the lower float engaged in the same second. The float is mounted slightly higher than the 30 percent line, about 4.5 centimetres, and because the controller accepts either the float or the ultrasonic threshold as permission to start, the float wins. This is the designed behaviour of two independent start paths, and the finding is that the effective lower threshold is set by a screw on the tank wall rather than by the configuration.

E4, switching frequency. Over the 84 minute session the pump started 18 times, 12.9 per hour, including starts commanded by hand during the command tests. The shortest gap between a stop and the next start was 32 seconds, above the 20 second minimum off time, so the relay never chattered. The level was trusted in 99.5 percent of messages, and no fault fired, which bounds the false alarm rate of all eleven rules at zero in 84 minutes of normal operation.

A fault found by this analysis. Separating automatic from manual starts exposed pump starts that belonged to neither: starts at 38, 46 and 51 percent with the device in automatic mode and no command sent. Each coincided with a device restart, identified by the sequence number returning to one, and in the replayed history the pump turned on in the second message after boot with the level reading exactly zero. The cause was an initialisation order defect. At start up the firmware sets the time of the last valid reading to the current time, so that the sensor timeout counts from power up; but the validity flag was derived from that same timestamp, so for the first seconds after every boot the level was declared valid while still holding its initial value of zero. The state machine saw an empty tank and started the pump, which then ran until the real level passed 70 percent. The fix requires at least one accepted reading before the level can be declared valid, and the status message now carries the reason for the last restart, so a restart caused by a supply dip can be told apart from a firmware upload. The fix was checked on the final firmware when it restarted at 20:20:52 on 5 October with the tank at 33 percent: the first level it published was 32.0 percent, never zero, and the pump stayed off until the draining level crossed 30 percent five seconds later, an ordinary automatic start. The demonstration repeats this check live by pressing the board’s reset button with the tank above 30 percent.

E8, control policy comparison. To show that the band and timing constraints matter, configuration A (49 and 51 percent, no timing constraints, approximating a single threshold) was compared with configuration B (30 and 70 percent with both timing constraints) on the simulator of Section 4.4, under the same ten minute continuous draw repeated three times, with one fixed target band from 25 to 85 percent for both. A switched 472 times per hour and B 8 times per hour, a reduction of 98.3 percent, with neither leaving the target band. The hardware session above, at 12.9 starts per hour with manual starts included and a different draw rate, is of the same order as the simulated figure for B.

5.4 Fault detection

E5, procedure. Each fault is injected by hand while `bench.py fault` is running; the operator presses Enter at the instant of the action, and the tool records the time to the first fault message, the fault code, the pump health label and the time until the pump state message reports the pump off. Table 5.4.1 lists the injections, the design delay and the measured result.

Table 5.4.1: E5 fault injection, firmware of 21 to 23 September.

Injection	Expected code	Design	Measured
Unplug the echo wire	SENSOR_TIMEOUT	45 s (90 s now)	40.2 to 41.0 s after the last trusted message, $n = 9$
Overflow condition reached	OVERFLOW	0.2 s	Pump off in the same control cycle, $n = 3$
Pump running, level not rising	NO_PROGRESS	90 s	Pump off in the same control cycle, $n = 15$; again $n = 2$ on 5 Oct
Disconnect one pump wire	NO_CURRENT	2 s	Not injected ; current path verified, see text
Manual start while a fault is active	rejected, fault_active	immediate	Rejected 4 of 4, ack in 31 to 43 ms

The last three measured rows come from the fault, pump event and command logs of 21 to 23 September rather than from a timestamped injection. They answer a different but equally important question: once a rule fires, how quickly does the pump actually stop? In all eighteen overflow and no progress events where the pump was still running, the message reporting the pump off was published in the same 200 millisecond cycle as the fault, within 0.12 seconds of it. Every manual start attempted while a fault was active was refused with the correct reason. The delay from the physical event to the fault was not timed with `bench.py fault`, and the group states plainly that `NO_CURRENT` has not been triggered by a physical injection: its 2 second delay is a design value, not a measurement. What has been measured is the signal the rule depends on. In the system test of 5 October the relay was closed by command for eight seconds and the current read 353 to 365 milliamperes in every message (44 millivolts at the pin, with the scale factor of the board’s 10 k Ω / 20 k Ω divider), and 0 in every message after it opened, so the present or absent decision the rule makes has a margin of about eleven times the 4 millivolt noise floor. The injection itself, pulling one pump wire during a fill, is part of the live demonstration.

The sensor timeout figure comes from the 23 September database, when the limit was 45 seconds, and is consistent with that design once the pipeline is accounted for: the level is marked untrusted 5 seconds after the last accepted reading, and the timeout counts 45 seconds from that same reading, so the fault arrives about 40 seconds after the last message that still showed a trusted level. Under earlier, shorter timeouts the same measurement gave 0.8 to 18 seconds, and the frequency of those faults during ordinary fills is how the group discovered that a short timeout was stopping normal operation. The final firmware extends the limit to 90 seconds while the level model is valid, for the blind zone reason given in Section 3.4; that value has not been re-timed by injection.

5.5 Latency and behaviour during an outage

E7, and why the headline number is the round trip. The group reports the command round trip as the primary latency result, not the one way telemetry delay. A one way delay is $t_{recv} - t_{send}$ read from two different clocks, so it carries their offset as a systematic error, and the network time protocol on a microcontroller over Wi-Fi is accurate only to tens of milliseconds, the same order as the quantity measured. Immediately after a reboot the group observed an apparent one way delay of about 1100 milliseconds while the round trip at the same moment was 187 milliseconds, direct evidence that the one way figure was reporting clock skew. The round trip is stamped by the server at both ends, so the offset cancels exactly.

A measurement defect found and corrected. The first implementation carried a device timestamp at one second resolution, whose quantisation error of up to 500 milliseconds exceeded the delay being measured: the median of 844.6 milliseconds over 372 samples was an artefact of the field width. A millisecond field was added alongside, and the same measurement then gave 245.0 milliseconds. The resolution of an instrument must be finer than the effect it is meant to resolve, otherwise the instrument measures itself.

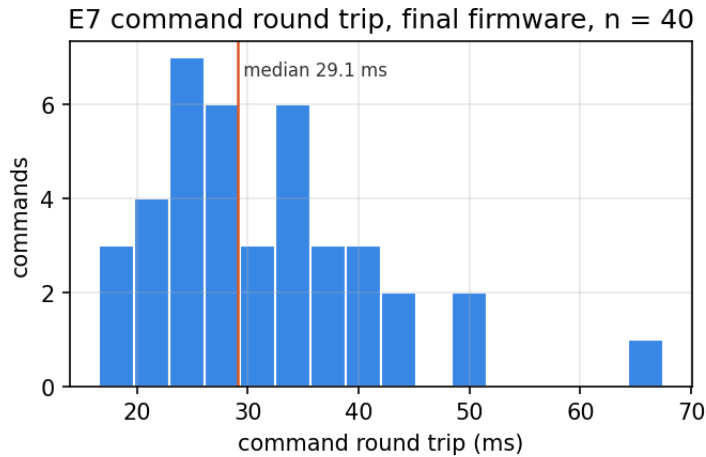


Figure 5.5.1: Distribution of command round trip times on the final firmware, 5 October: 40 mode commands sent one second apart by the system test.

Table 5.5.1: Latency and outage results.

Quantity	Result
E7: command round trip, final firmware, 5 Oct, median, 95th percentile, maximum (ms) (result of record)	29.1, 48.5 and 67.5, $n = 40$, none lost
E7: command round trip, 23 Sep session (ms)	42.4, 137.8 and 181.6, $n = 75$
E7: telemetry one way, 23 Sep, median and 95th percentile (ms)	166.9 and 238.0, $n = 4826$, comparison only
E7: command round trip, first firmware (ms)	225.8 and 575.3, $n = 14$
E7: telemetry one way at second resolution (ms)	844.6, $n = 372$, quantisation artefact
E6, final firmware: outage, device traffic dropped	120.0 s, pump running, level 9.0% at the cut
E6: pump behaviour during the outage	kept filling to 50.9%, no transition; stopped itself at 70.7%
E6: last will published by the broker	21.8 s after the cut
E6: device back online after restore	10.0 s; first live message after 10.9 s
E6: messages buffered and replayed	125, sequence 707 to 831, buffer never full
E6: messages lost	4 of 129, sequence 703 to 706
E6: longest control period during the outage	0.88 s
E6, first firmware: 60 s outage, pump off	32 replayed, 10 lost

The round trip fell from a median of 226 milliseconds on the first firmware to 42 on 23 September and 29 on the final firmware. Section 5.6 accounts for most of it, the removal of Wi-Fi power saving. The later sessions also ran on different access points, and the group has not separated the remaining improvement between the two causes.

E6, the outage procedure. The outage test isolates the device alone, leaving the broker and the backend connected to each other, because that is the failure the requirement describes. `bench.py outage` either drops the device’s traffic at the laptop firewall for a fixed interval or prompts the operator to switch the access point off and on, records every MQTT message, and reports the time until the broker published the last will, the time to reconnect, the messages replayed and lost, every pump transition contained in the replayed history, and the longest control period during the outage. An earlier attempt stopped the broker process instead; that is not equivalent, because the device reconnected one second before the backend resubscribed and the whole replayed batch was published to a broker with no subscriber, so none of it reached storage.

Reading the outage result. The test was repeated on the final firmware on 5 October with the

pump running at the moment of the cut, the case the requirement is really about. Figure 5.5.2 shows it. For 120 seconds the device could not reach the broker; the backend and the broker stayed up and kept each other connected. The device kept filling, from 9.0 to 50.9 percent, without a single pump transition, crossed the blind zone on its model after the link returned, and stopped the pump itself at 70.7 percent. The broker published the last will 21.8 seconds after the cut, which is the one and a half keep alive intervals the MQTT specification allows for a 15 second keep alive [6]. Ten seconds after the link was restored the device was online again, and the 125 messages it had buffered were replayed in order.

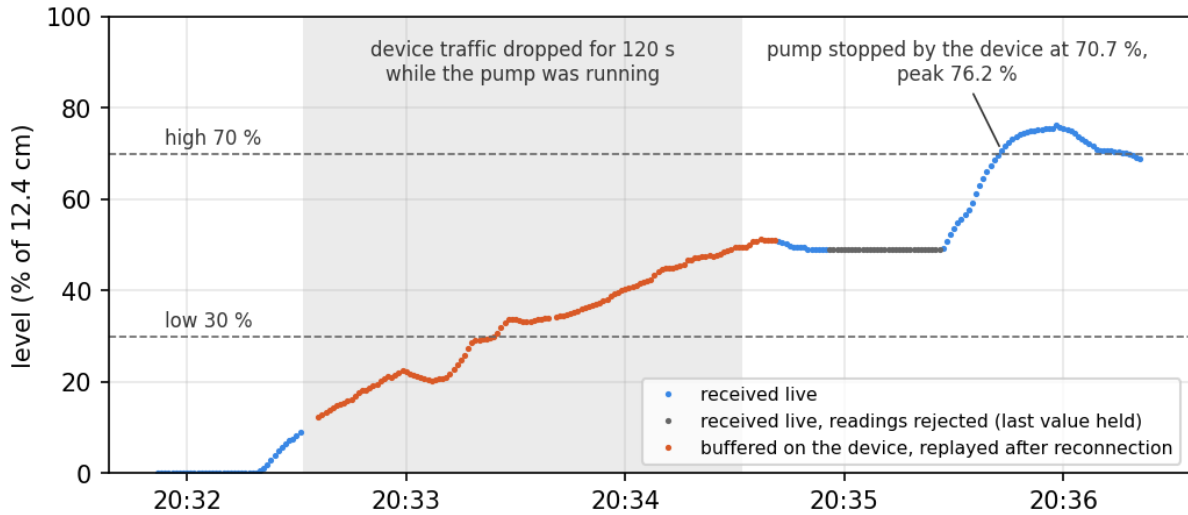


Figure 5.5.2: E6 on the final firmware, 5 October. Grey band: device traffic dropped for 120 s while the pump was running. Orange: samples buffered on the device and replayed after reconnection. Grey points: readings rejected in the blind zone while the model bridged the gap.

The four lost messages deserve more attention than the 125 that survived. They were published into a socket the client still believed open: the buffer protects only samples generated after the firmware has decided the link is down. On the first firmware that decision waited for the keep alive and lost ten messages; the final firmware closes a socket that has refused data for three seconds, and the loss fell to four, one per second of the detection window. The residual gap is bounded by that window, not by the buffer size. The buffer of 240 entries was half full after a two minute outage, so about four minutes can be covered. The longest control period during the outage was 0.88 seconds, against three to ten seconds per reconnection attempt on the first firmware, which supports the claim of Section 2.2 that the network never sits on the path of a pump decision.

Automated system test. The tool `tools/systest.py` repeats the acceptance checks without an operator; Table 5.5.2 lists its result on the final firmware.

Table 5.5.2: Automated system test on the final firmware, 5 October, 20:37.

Test	Check	Result
T1 connectivity	10 s of telemetry: rate, sequence gaps, fields, retained status	Pass: 10 messages, no gap
T2 security	Viewing open; command without session and wrong password refused	Pass: 200, 401, 401
T3 commands	40 mode commands, round trip, none un-acknowledged	Pass: median 29.1 ms
T4 rejections	Pump in automatic mode; restart inside the off time; clear without fault	Pass: three correct reasons
T5 manual pump	Current with the relay closed and open	Pass: 353 to 365 mA, then 0

5.6 Reducing transport latency at the radio

E9. The spread of the first round trip figures was wider than a local wireless network should produce, with a 95th percentile near 900 milliseconds at a signal strength better than -20 dBm. The cause was the radio: the Arduino port for the ESP32 enables modem sleep by default, so the receiver is powered down between beacon intervals and a packet waits for the next scheduled wake. Disabling it is a single call after association. The experiment held everything else fixed.

Table 5.6.1: Command round trip before and after disabling Wi-Fi power saving.

Statistic	Power saving on	Power saving off	Change
Minimum (ms)	144	29	-80%
Median (ms)	462	187	-60%
95th percentile (ms)	884	315	-64%
Maximum (ms)	906	402	-56%
Samples	12	12	

The improvement is largest in the tail, the signature of a scheduling effect rather than a bandwidth effect. The cost is roughly double the radio’s continuous current, immaterial for a mains powered controller and decisive for a battery powered one; a latency figure quoted without its power cost is not an engineering result. The change does not alter the 200 millisecond control period, and therefore does not change how quickly the tank reacts to its own level.

5.7 Limitations

The group believes that pointing precisely at where the design falls short carries as much engineering value as presenting what worked.

On hardware, the most serious limitation is the flow measurement, and the group words it carefully because the assignment makes a flow sensor mandatory. The requirement is implemented in hardware: two sensors are fitted, connected through dividers, read by interrupt and published. Validation is partial. Until 22 September the pump's conducted interference was forty times the real pulse rate; after the rewiring of 5 October the inlet sensor counts water and has been calibrated against the ruler and the tank geometry on one run, while the outlet sensor stays below its starting flow at bench drain rates. Control and volume accounting still use flow derived from the level, so the inlet is a calibrated constant in the volume counter rather than an independent observation. Three consequences follow and are stated rather than hidden: the volume error of -5.2 percent with a 23 point spread is a property of that constant, and it grows when the pump delivery drifts, as it did to 0.24 litres per minute on 5 October; the literal rule "pump on with near zero flow" is carried by level progress instead of the turbine; and the edge leak rule is inactive. The next steps are to move the volume counter to the inlet sensor after validating it over complete cycles, and to replace the outlet sensor with one whose range starts near 0.05 litres per minute.

Second, the absolute current is not verified. The firmware first assumed a one to one divider on the current sensor while the board, as drawn in Figure 2.4.1, uses 10 k Ω and 20 k Ω ; the scale factor has been corrected to the schematic, which turns the 44 millivolts read with the pump running into 353 to 365 milliamperes. That is still above the 100 to 200 milliamperes the pump is rated for, so one reading with a multimeter in series is needed. The binary use of current, present or absent, which is all the fault rules and the health classifier need, is unaffected. Third, the ultrasonic sensor has a blind zone in this tank: with the surface 7.9 to 8.7 centimetres below the transducer it returns a stable stray echo instead of the water. The firmware rejects those readings and crosses the band on its model, which costs about one minute without a measured level in every fill and adds overshoot when the filter catches up, as in Section 5.3. The beam is also wider than the tank, so a share of pings return from the wall or the floor. A stilling tube, foam around the transducer or a wider vessel would remove both causes. Fourth, the sensor can lock up after a supply disturbance and recover only when its supply is cycled; powering it from a controllable output would let the firmware do that by itself.

On measurement, the headline latency no longer relies on two clocks, as Section 5.5 explains; the residual limitation is that the clock synchronisation error was characterised only by observation. The outage buffer protects only samples generated after the firmware decides the link is down, so about three seconds are lost at the start of every outage; and an outage that also restarts the broker discards the replayed batch, since the device can flush before the backend resubscribes. Repeat counts of six to seventy five suffice for means and spreads but not for formal hypothesis tests. The rolling baseline needs at least three observations of each half hour slot before it may alarm, and after four bench days, held at different hours, no slot has been seen on three days, so its detection performance is unmeasured. Of the fault rules, `NO_CURRENT` was never triggered by a physical injection, and the fault delays of Table 5.4.1 were

measured on the September firmware and not re-timed on the final one.

On operations, the public tunnel address changes every time it is started, which suits a demonstration but not a deployment, where a named tunnel or a fixed host is needed; administrator sessions are held in memory and do not survive a restart; and there is one administrator password rather than per user accounts. The broker traffic is not encrypted inside the laboratory network, so the system is not secured end to end by TLS, as Section 4.6 states. There is no remote firmware update.

On the interface, a defect that survived three separate inspections is worth recording because of how it hid rather than because of its size. The query that feeds the history charts applied its row limit after ordering the window in ascending time, so it returned the oldest rows of the window rather than the newest. At the specified rate of one message per second a ten minute window holds six hundred rows and never reaches the two thousand row limit, so the fault was invisible in every ordinary session. Running the simulator at ten times real time raises the rate to roughly five messages per second, the window then holds close to three thousand rows, and the charts silently froze in the past while the live status panel beside them continued to update. Nothing failed and nothing was logged; the two halves of the same screen simply disagreed.

The general lesson is that a pagination limit at the query layer must truncate from the end that is being discarded, not from the end being displayed. Ordering ascending and then limiting is correct only while the result set is known to fit, which makes it a latent fault that waits for the data rate to rise. The corrected query selects the newest rows first and reorders them for plotting. The group notes that the defect was found only because the acceleration factor of the simulator pushed the system past a threshold the specified rate never approaches, which is an argument for exercising a system deliberately outside its nominal operating point rather than only at it.

Future directions include completing the flow measurement as described above, which would make the inlet an independent observation and let the turbine and level estimates be checked against each other as a leak test in their own right; MQTTS on port 8883; a Kalman filter in place of the alpha-beta filter, using the pump state as a known input; continuous speed control by pulse width modulation; and multiple nodes feeding one backend.

5.8 Conclusion

The project delivers a working closed loop water tank that measures its level, controls its pump, detects and classifies faults, keeps working without the network, and can be watched from anywhere on the Internet while only an administrator can operate it. Every line of firmware, backend and interface was written by the group, and every figure in this chapter can be regenerated from the database by the scripts in the repository.

Three decisions carry the design. First, the whole control law and every safety rule live in the edge controller, derived from the rule that anything needing an immediate safe stop belongs at the edge; the outage requirement then follows from the architecture instead of being added to it. Second, safety is structural: every path that can start the pump passes through one guard function that is re-evaluated

every cycle, and overflow is guarded by independent layers of which the raw distance guard, the float and the run limits do not depend on the filtered level at all. Third, the group trusted measurements over datasheets and over its own expectations: the pump delivers a fifth of its rated flow and not even that consistently, the flow sensors counted interference rather than water until the power stage was rewired, the level sensor needs a fifteen sample robust filter in this tank and has a blind zone that only a stepwise calibration revealed, and the analysis of the control session found an initialisation defect that started the pump after every restart. Each of these was found by counting, and each is fixed or stated.

On the bench the controller stopped every automatic fill of the control session between 70.1 and 70.8 percent with at most 1.85 percent of overshoot, switched 12.9 times per hour with no chatter, raised no false alarm in 84 minutes and estimated pumped volume to -5.2 percent on average. Measured against a ruler and the pumped volume, the level is within 0.68 centimetres root mean square outside one blind zone. On the final firmware it kept filling through a two minute network outage and stopped itself at 70.7 percent, lost 4 of 129 messages, and answered commands with a median round trip of 29 milliseconds. The open items are stated rather than hidden: the flow sensors are fitted and the inlet is calibrated but not yet used for accounting, the no current rule has not been triggered by injection, and the MQTT path is not encrypted.

5.9 Member contributions

Table 5.9.1: Distribution of work among group members.

Member	Area of responsibility
Trịnh Trần Quốc Bảo 24119016	Problem statement and requirements; system architecture, the edge decision principle, the bench rig and signal conditioning. Presents Parts 1 and 2.
Đặng Cao Dương 24119024	Lessons from bench faults, limitations and next steps, conclusion; coordinates the live demonstration and explains each test case and its purpose. Presents Part 7 and the demo.
Huỳnh Lê Thành Nhân 24119068	Firmware and code: MQTT protocol and acknowledgements, level filter pipeline, flow from level, state machine, fault rules, safety guard, pump health and leak detection. Presents Parts 3 and 4.
Nguyễn Đoàn Hoàng Ngân 24119064	Backend and SQLite schema, REST API, dashboard and Cloudflare tunnel, security model, ring buffer and outage recovery. Presents Part 5.
Tô Ngọc Thảo Uyên 24119098	Experiment plan and analysis: control response, volume estimation error, latency and Wi-Fi power saving, fault and outage results. Presents Part 6.

The assignment requires every member to present and answer technical questions, so the table records who leads each part of the work and of the presentation rather than limiting anyone’s scope of understanding.

Deliverables are the working physical prototype; the firmware with its per-component test programs; the backend, dashboard and simulator; the protocol specification; the test procedures and analysis scripts (`docs/TESTING.md`, `tools/`); a code guide for newcomers (`docs/CODE_GUIDE.md`); the raw database; the presentation slides; and a backup demonstration video.

Appendix A

BENCH FAULT LOG

The faults recorded here were all found after the prototype was assembled and nominally working. The group presents them because each was invisible from the code alone and each was resolved by a measurement rather than by argument.

A.1 An inverted relay made the firmware believe the pump was off

The most serious fault was also the simplest. The configuration declared the relay module active low, so the firmware drove the control pin high to stop the pump. The module fitted was active high. Every command to stop therefore started the pump, and the telemetry reported `pump: false` throughout. The tank filled from 0.2 to 11.2 centimetres and came close to overflowing while the interface displayed a stopped pump and an overflow lock.

The fault also corrupted the current measurement, in a way that concealed it. The quiescent offset of the Hall sensor is tracked by an exponential average that updates only while the firmware believes the pump is off. Because the firmware believed that permanently, the average absorbed the running current of the pump as its own zero, and the reported current settled near zero while the motor ran.

Determining the true polarity required care, because the obvious method gives the wrong answer. The Hall current sensor is bidirectional: its output rests at mid scale and moves up or down depending on the direction of current through the primary conductor. A larger reading therefore does not imply a larger current, and the first version of the group's test concluded the opposite of the truth on exactly that basis. The quantity that does separate the two states is the *ripple*, because a brushed motor draws a visibly rough current through its commutator while an open relay draws a flat one. Measured over two rounds, the control pin low gave a mean of 1545.88 millivolts with 4.47 millivolts of ripple, and the pin high gave 1516.54 millivolts with 27.05 millivolts of ripple. The ripple ratio of 6.0 identifies the energised state unambiguously and does not depend on which way the load is wired, whereas the means differ by only 29 millivolts and in the misleading direction.

A.2 An automatic direction level shifter consumed a third of the echoes

The echo line of the ultrasonic sensor was initially routed through a TXS0108E translator. That part is intended for bidirectional open drain buses: it carries integrated pull up resistors of about 10 kilohms on both ports and a one shot circuit that accelerates edges. The echo output of an HC-SR04 is a strong push pull driver, and a pulse width is the only carrier of information in the signal, so both mechanisms work against it. Replacing the translator with a two resistor divider on the echo line alone changed the miss rate from between 28 and 30 percent of pings to zero in 179 consecutive samples.

A.3 Fifty hertz on a floating signal line reads as plausible flow

Both flow inputs reported between 0.26 and 0.51 litres per minute with the pump stopped. Converting the reading back to its underlying pulse rate showed exactly 50.00 hertz on one channel, unchanging across every sample. The control experiment was to close the drain valve, after which no flow of any kind was physically possible; the channel continued to report 50.00 hertz. A signal locked to the mains frequency and indifferent to whether water can flow is induced interference, not measurement.

The cause is impedance. The internal pull up of the microcontroller is about 45 kilohms, which leaves the line high enough in impedance to act as an antenna. The specified remedy, an external 4.7 kilohm pull up to the 3.3 volt rail, lowers the source impedance by an order of magnitude. A second and larger disturbance appears only while the pump runs, when both channels rise together to about 4750 hertz, a figure that would correspond to 48 litres per minute against a measured pump delivery of 0.36. Two independent sensors agreeing to within 0.7 percent are not measuring water; they are sharing a common mode disturbance injected by the motor.

A.4 A silently truncated message stopped only the database

Adding two validity fields to the telemetry payload pushed its length past the 448 byte buffer that served it. The JSON library truncates without raising an error and still returns a string, so the device published well formed nonsense. On the server the parse exception was raised inside a callback whose exceptions the client library discards. The result was a system in which the device published once per second, the broker delivered every message, the health endpoint reported the service as connected, and the database received nothing at all for several minutes while the dashboard displayed a frozen value.

The lesson the group takes is that a serialiser which truncates silently and a callback which swallows exceptions are individually defensible and jointly produce a failure with no symptom at the point of failure. Both ends now report the condition explicitly, and the buffer carries margin above the maximum payload rather than matching it.

A.5 A command that worked but was never confirmed

Manual pump commands were executed by the device, yet the command log kept them as pending indefinitely, while mode commands were confirmed normally. The broker log showed the difference: acknowledgements of mode commands were 118 bytes long, acknowledgements of pump commands exactly 256 bytes, the size of the acknowledgement buffer.

The cause was an interaction between two libraries. The MQTT client delivers an incoming message as a pointer into its internal buffer, and the same buffer is reused for outgoing messages. The JSON parser, handed a non-constant pointer, parses in place and returns string references into that buffer rather than copies. A pump command starts the pump, which publishes the new pump state before the acknowledgement is sent; that publish overwrote the buffer, so the command identifier being echoed back had already become arbitrary bytes. Mode commands publish nothing in between and were unaffected. The fix copies the incoming message into a buffer owned by the handler before parsing. The general point is that a zero-copy parser makes the lifetime of the input part of the program's correctness, and that lifetime is decided by another library.

A.6 A robust filter gated by a fragile statistic

After the level filter was widened to a fifteen sample median, the group counted, for every ping, which validation stage rejected it. In one ordinary fill of 605 pings, 244 were rejected, and 98 percent of those rejections came from a single stage: the check that the spread of the window stayed below a limit. Missing echoes accounted for only 2 percent. The sensor was answering almost every time.

The spread had been measured as maximum minus minimum. That statistic is controlled entirely by the single most extreme sample, so one stray echo from the tank floor among fifteen good ones was enough to discard the whole window. The median was chosen precisely because it tolerates isolated outliers; gating it with the statistic most sensitive to isolated outliers undid that choice. Replacing the range with the interquartile range, which ignores the top and bottom quarter of the window, raised the acceptance rate from 60 to 93 percent in otherwise identical conditions. The interquartile range still grows when the sensor genuinely alternates between the water surface and the floor, because then roughly half the samples sit in each group, which is exactly the case the check exists to reject.

A.7 The level sensor is excellent, and unusable, in the same installation

With the water still, twenty consecutive samples read between 3.87 and 3.89 centimetres, a standard deviation of 0.010 centimetres. With the pump running, the same sensor collapses to reporting an empty tank while the tank is demonstrably filling. The group verified that the validity gates were not the cause by disabling them: the raw reading itself falls to the empty value, so the gates were reporting the data correctly rather than rejecting it wrongly.

This pair of results is the clearest argument in the project for layered protection. Throughout the interval in which the level reading was false and the validity flag had been forced true, the rule that stops the pump when the level fails to rise still fired and still stopped it. That rule consults the clock and the level history, never the validity flag, which is precisely why it survived.

A.8 A repeatable echo is not necessarily the water

For several days the level stalled at 55 to 60 percent in most fills, the sensor was reported lost, and the pump stopped short of the target. The cause appeared only in the stepwise calibration of 5 October. With the surface 7.9 to 8.7 centimetres below the transducer, all forty pings at a step returned the same wrong value: a level of 1.01 centimetres for a true 7.21, or a distance beyond the floor for a true 7.97. Above and below that band the same sensor read within a centimetre.

The episode overturned a rule the group had added a day earlier: a reading that the model gate rejected was accepted anyway if it repeated for several cycles, on the reasoning that noise does not repeat and the water does. The stray echo repeated perfectly, the rule adopted it as an empty tank, and the pump started about every forty seconds. Consistency over time is evidence of a physical reflector, not of the right one. The rule was removed; the firmware now discards distances beyond the floor, crosses the band on a model fed by the measured inflow, and allows that model up to 90 seconds.

A.9 The rewiring, not the diode, removed the flow interference

Section 5.2 records that a flyback diode and moving the wires apart left the 1500 hertz pump interference on the flow inputs unchanged, which placed it in the shared supply and ground. On 5 October the power stage was rebuilt: the relay now switches the supply ahead of the current sensor and the pump, a diode and a capacitor sit at the motor leads, a 470 microfarad capacitor buffers the controller board, and the two flow inputs were swapped between their pins. After the rebuild the inlet channel read 0 hertz with the pump off and 20 to 30 hertz with it running, the first time in the project it counted water. The group did not change one thing at a time on that day, so it cannot say which change removed the interference; it records the combination that works, and the earlier measurements that ruled out radiation as the path.

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